

FYP Document “CLARITYWORKS”

Software Requirements and Design Specification

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1 Introduction

1.1 System Introduction

ClarityWorks is a web-based platform designed to enhance content accessibility by integrating text readability analysis with controlled rewrites. It serves as a comprehensive solution where content creators, educators, and students can analyze text complexity, identify problem areas, generate simplified versions, and track improvement over time, all under administrative oversight. Key features include multi-metric readability analysis, difficult word and sentence identification, controlled rewrites targeting specific grade levels, before-and-after comparison, and a progress tracking system. ClarityWorks aims to foster accessible and clear communication, combining analytical tools with artificial intelligence to help writers reach their intended audiences effectively.

1.2 Background of the System

In educational institutions, publishing organizations, and content creation industries, writers often struggle to determine whether their text is appropriate for their target audience. Traditional readability assessment methods are time-consuming, require specialized knowledge, and depend on manual calculations. Many content creators lack systematic tools to identify complex vocabulary, lengthy sentences, or difficult grammatical structures that hinder comprehension. ClarityWorks was conceived to address these gaps by providing a centralized platform for text analysis, readability assessment, and controlled rewrites. The system integrates with AI services and natural language processing libraries, allowing users to easily submit text, receive instant feedback, and generate simplified versions that maintain original meaning while improving accessibility.

1.3 Objectives of the System

The main objectives of ClarityWorks are as follows:

- **Centralized Platform:** Develop a unified, web-based system to support users and administrators in analyzing and improving text readability.
- **Comprehensive Analysis:** Provide multiple readability metrics including Flesch Reading Ease, Flesch-Kincaid Grade Level, SMOG Index, and Automated Readability Index to assess text complexity.
- **Problem Identification:** Automatically identify and highlight difficult words and complex sentences that reduce comprehension.

- **Controlled Simplification:** Enable users to generate simplified text versions targeting specific grade levels (Grades 3-12, College) using rules and constraints.
- **Comparison Tools:** Implement before-and-after comparison views to quantify readability improvements and demonstrate simplification effectiveness.
- **Role-Based Access:** Implement secure authentication and role-based dashboards for users and administrators.
- **Administrative Control:** Facilitate user management, analysis monitoring, and system oversight through an admin console.
- **History Tracking:** Maintain analysis history allowing users to review past assessments and track improvement over time.

1.4 Significance of the System

ClarityWorks provides significant convenience by enabling instant text analysis, automated problem identification, and controlled rewrites. Users no longer need to manually calculate readability formulas or spend hours rewriting content, as the platform offers complete transparency regarding text complexity, specific issues, and improvement recommendations. The before-and-after comparison feature demonstrates measurable improvements, contributing to a more efficient and productive writing process. It supports better audience targeting by allowing creators to adjust content for specific reading levels through a centralized interface.

2 Overall Description

2.1 Product Perspective

ClarityWorks operates at the intersection of content analysis tools and writing assistance platforms, distinguishing itself by focusing on readability assessment and audience-appropriate simplification. Unlike generic grammar checkers or standalone AI tools, ClarityWorks integrates comprehensive readability analysis with intelligent simplification, offering functionalities like multi-metric scoring, problem identification, grade-level targeting, and before-after comparison. It leverages technologies such as web connectivity and ML-based processing, innovating in how they are applied to content accessibility.

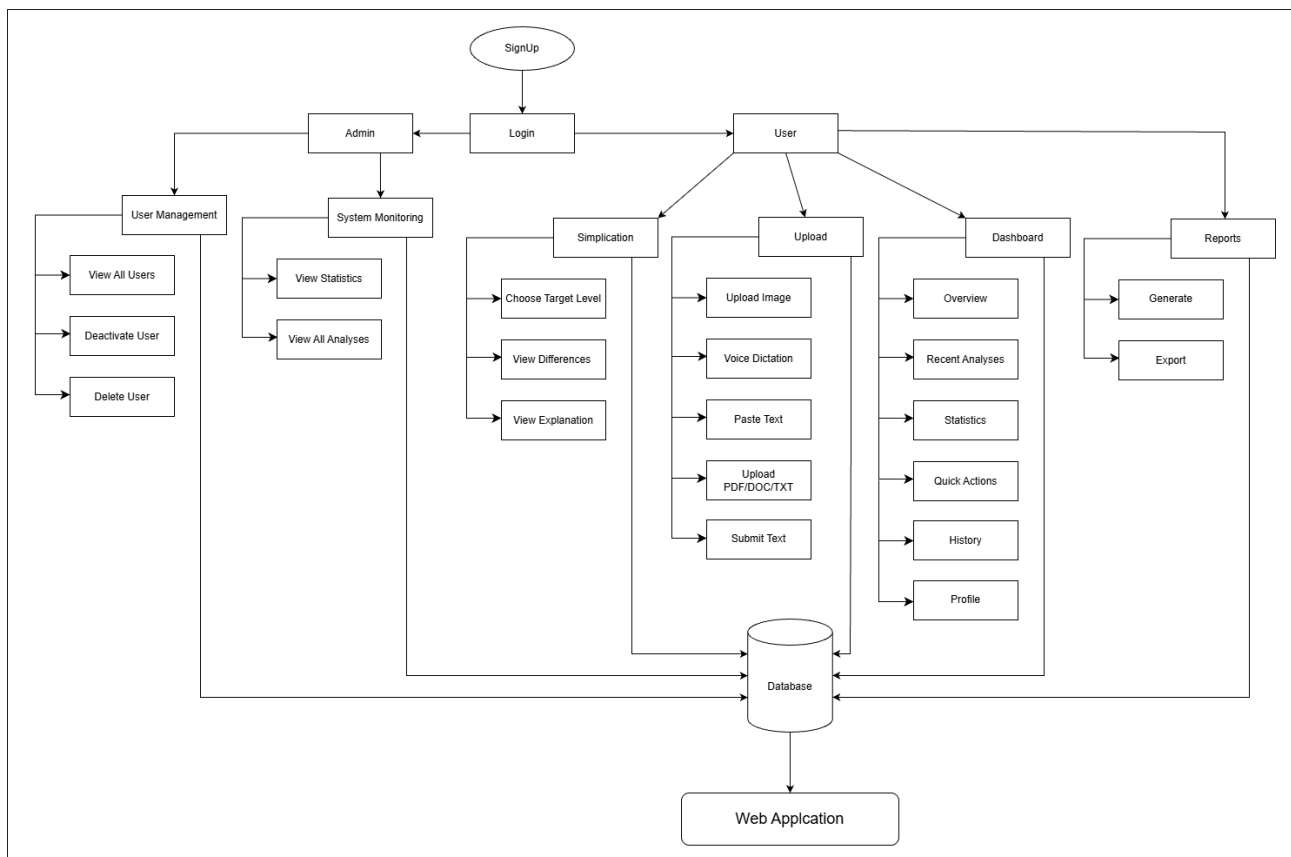


Figure 2.1: System Design

2.2 Product Scope

ClarityWorks is designed to be a comprehensive platform for content creators, aiming to improve how they assess and enhance text accessibility. Its scope includes features for analyzing text (calculating readability metrics, identifying difficult words and sentences), simplifying content (controlled rewrites for specific grade levels), comparing results (side-by-side original and simplified versions), and exporting reports. The system also includes

administrative oversight to monitor user activities and ensure a reliable environment. ClarityWorks seeks to bridge the gap between complex writing and accessible communication, creating a productive environment where creators can optimize content for diverse audiences.

2.3 Product Functionality

2.3.1 Analysis Tools

- **Readability Metrics:** Users can calculate Flesch Reading Ease, Flesch-Kincaid Grade Level, SMOG Index, and Automated Readability Index to assess text complexity.
- **Problem Identification:** Users can identify and view difficult words and complex sentences that hinder comprehension.

2.3.2 Simplification Features

- **Grade Level Selection:** Users can specify target reading levels (Grade 3-12, College) before requesting simplification.
- **Controlled Rewriting:** Users can generate simplified versions that preserve meaning while improving readability.
- **Comparison View:** Users can view original and simplified text side-by-side with highlighted changes and improvement metrics.

2.3.3 Admin Console

- **Administrative Monitoring:** Admins can monitor user activities, view system statistics, and manage user accounts to ensure proper usage and maintain a secure environment.

2.4 Users and Characteristics

- **User:**
 - **Description:** Primary users who use the platform to analyze text, identify issues, and generate simplified versions. They vary in writing expertise, technical knowledge, and content types.
 - **Activities:** Registering, logging in, submitting text for analysis, viewing readability metrics, identifying problem areas, generating simplified versions, comparing results, tracking progress, exporting reports, and managing profile settings.
- **Administrator:**
 - **Description:** Oversees the platform's functionality, manages users, monitors system activity, and ensures a secure environment.
 - **Activities:** Monitoring user activities, managing user accounts, viewing system statistics, and addressing technical issues.

2.5 Operating Environment

ClarityWorks is a web-based application developed using React with TypeScript for the frontend and Python with Flask for the backend, with PostgreSQL as the database.

2.5.1 Hardware Platform

Designed to run on desktops, laptops, and tablets via modern web browsers.

2.5.2 Minimum Hardware Requirements

- **Processor:** Dual-core 2.0 GHz or higher
- **RAM:** Minimum 2 GB for mobile devices, 4 GB for desktops/laptops
- **Storage:** At least 200 MB of free space for browser cache and local data

2.5.3 Software Requirements

- **Web Browsers:** Latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari
- **Operating System:**
 - Windows:** Windows 8 or higher
 - macOS:** macOS 10.13 (High Sierra) or higher

3 Specific Requirements

3.1 Functional Requirements

3.1.1 Authentication & User Access

Table 3.1: Functional Requirements for Authentication & User Access

Requirement ID	Requirement	Description
CFR-001	Registration and Authentication	Users must be able to securely register. The system must validate email uniqueness and enforce password requirements.
CFR-002	Secure Login and Role-Based Access	The system must be able to securely authenticate users and authorize access based on roles.
CFR-003	Logout	The system must allow users to log out, terminating their session and clearing all authentication tokens.

3.1.2 Profile & User Management

Table 3.2: Functional Requirements for Profile & User Management

Requirement ID	Requirement	Description
CFR-004	User Profile Management	Users must be able to manage their profiles, update name, profile picture and email, and change password with current password verification.
CFR-005	Admin User Management	Authorized admins must be able to view all user accounts, search users, deactivate/reactivate accounts, and permanently delete users.

3.1.3 Text Input & Processing

Table 3.3: Functional Requirements for Text Input & Processing

Requirement ID	Requirement	Description
CFR-006	Multiple Text Input Methods	The system must accept text through direct paste, document upload, microphone dictation, and image upload.

CFR-007	Text Validation and Preprocessing	The system must validate text, display real-time character/word count, and remove excessive whitespace.
CFR-008	Document Text Extraction	The system must extract text content from uploaded documents and images, validate file formats, and display extracted text for user review before analysis.

3.1.4 Readability Analysis

Table 3.4: Functional Requirements for Readability Analysis

Requirement ID	Requirement	Description
CFR-009	Readability Metrics Calculation	The system must calculate multiple readability metrics against the submitted text.
CFR-010	Difficult Word Identification	The system must identify and highlight uncommon words, provide simpler alternatives, and calculate the percentage of difficult words in the text.
CFR-011	Complex Sentence Analysis	The system must identify and highlight sentences with many words or multiple clauses, assign complexity scores, and provide simplification suggestions.
CFR-012	Analysis Results Display	The system must display all metrics with visual representations, provide overall readability assessment, and save results to database.

3.1.5 Controlled Text Simplification

Table 3.5: Functional Requirements for Controlled Text Simplification

Requirement ID	Requirement	Description
CFR-013	Target Level Selection	The system must allow users to select target levels for simplification and only permit simplification to lower levels than current text.
CFR-014	Text Simplification	The system must apply controlled simplification to replace difficult words, break long sentences, and reduce complexity while maintaining meaning.
CFR-015	Simplification Comparison	The system must display original and simplified text side-by-side with highlighted changes, show before/after metrics comparison, and allow users to manually edit or revert changes.

3.1.6 History Management & Export

Table 3.6: Functional Requirements for History Management & Export

Requirement ID	Requirement	Description
CFR-016	User History Management	The system must save all text analyses with original text, metrics, and simplified versions to user's history. Analyses can be deleted with confirmation.
CFR-017	Admin History Management	Authorized admins must be able to view all analyses, filter analyses, and permanently delete analyses.
CFR-018	Report Export	The system must allow users to export analysis results as PDF documents including original text, metrics, difficult words/sentences, simplified text and comparison charts.

3.1.7 User Dashboard

Table 3.7: Functional Requirements for User Dashboard

Requirement ID	Requirement	Description
CFR-019	Dashboard Overview	The system must display a personalized dashboard showing summary statistics recent analyses, usage metrics, and quick action buttons.

3.1.8 Admin System Statistics

Table 3.8: Functional Requirements for Admin System Statistics

Requirement ID	Requirement	Description
CFR-020	System Statistics Dashboard	The system must provide admins with statistics including total users, total analyses, analyses per user, and usage trends over time.

3.2 Behaviour Requirements

3.2.1 Use Case Diagram for ClarityWorks

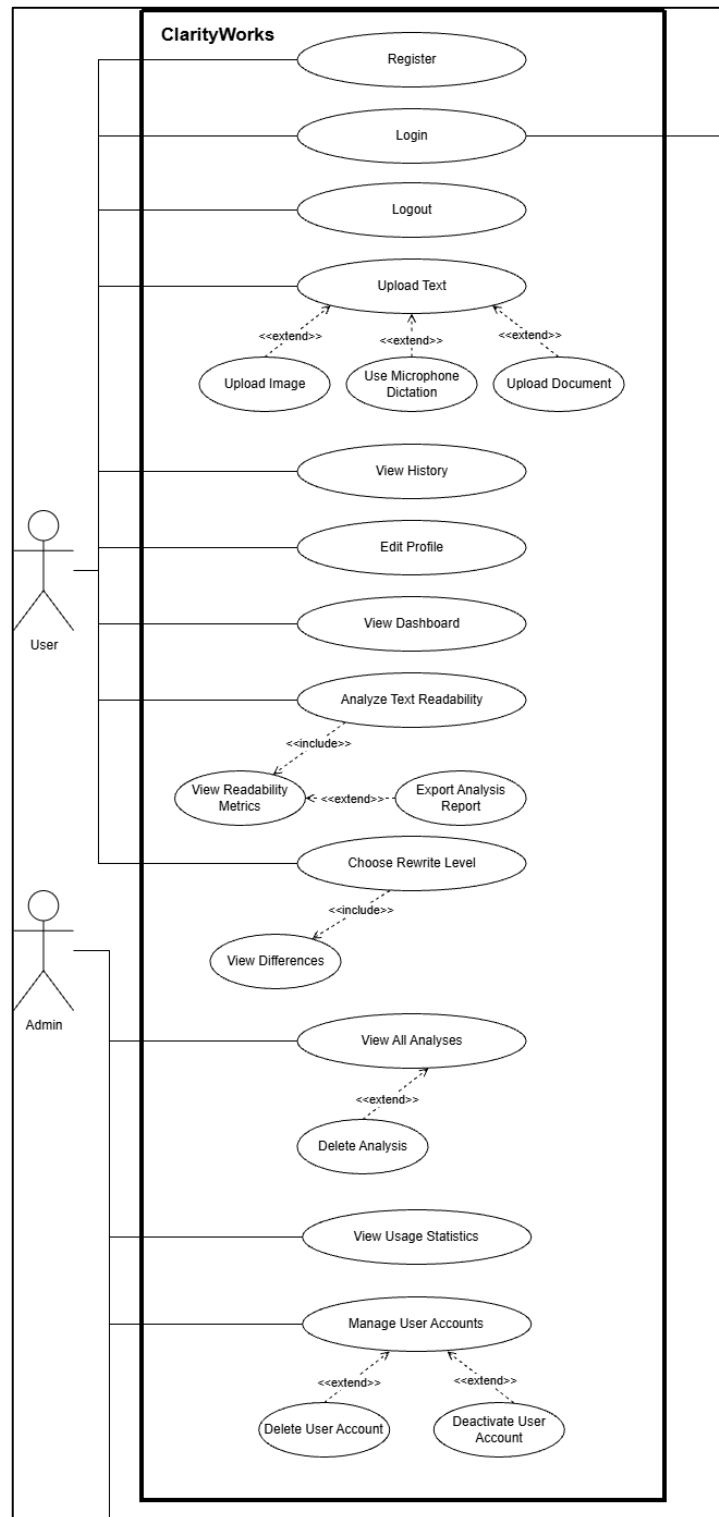


Figure 3.1: Use Case Diagram of System

3.2.2 Use Case #1: Register



Figure 3.2: Use Case Diagram for Register

Table 3.9: Use Case for Register

Field	Details
Actors	User
Description	User creates a new account in the system.
Pre-Condition	User is not logged in or registered.
Steps	1. Navigate to registration page. 2. Fill in required information. 3. Submit the form. 4. System validates and creates account.
Post-Condition	Account is created, and user can log in.

3.2.3 Use Case #2: Login

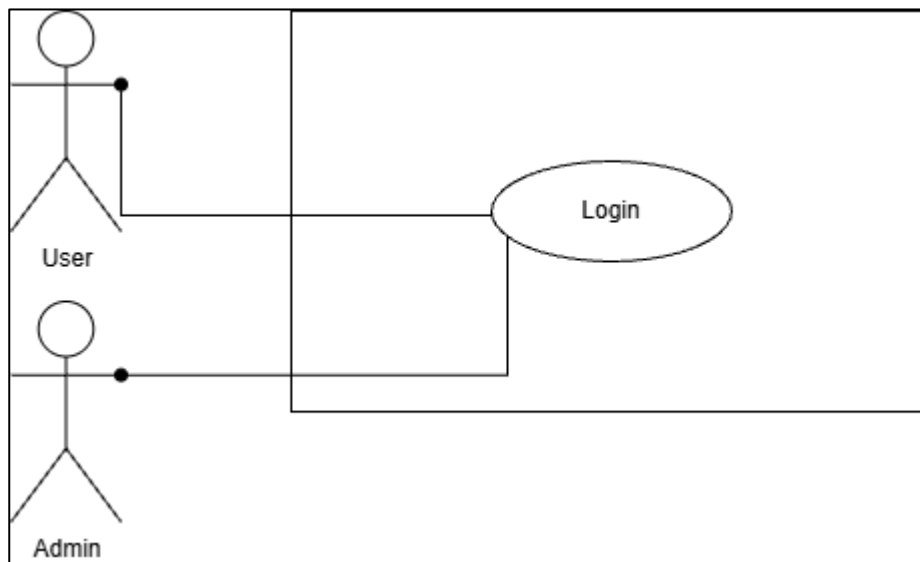


Figure 3.3: Use Case Diagram for Login

Table 3.10: Use Case for Login

Field	Details
Actors	User, Admin
Description	Authenticate and establish a user session.
Pre-Condition	User/Admin must have a registered account.
Steps	1. User/Admin enters username/email and password. 2. User/Admin clicks login. 3. System verifies credentials. 4. System creates session.
Post-Condition	User is logged in and redirected to dashboard/admin panel.

3.2.4 Use Case #3: Upload Text

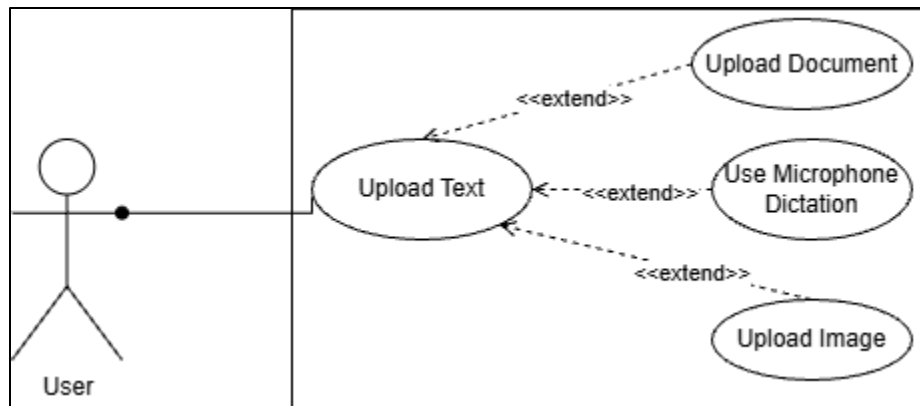


Figure 3.4: Use Case Diagram for Upload Text

Table 3.11: Use Case for Upload Text

Field	Details
Actors	User
Description	User uploads text from various sources for analysis.
Pre-Condition	User must be logged in.
Steps	1. User chooses upload method. 2. User uploads raw text, document file, uses microphone dictation, or uploads image. 3. System prepares text for analysis.
Post-Condition	Text is extracted and ready for analysis.
Relationships	Extends: Upload Document, Use Microphone Dictation, Upload Image

3.2.5 Use Case #4: View History

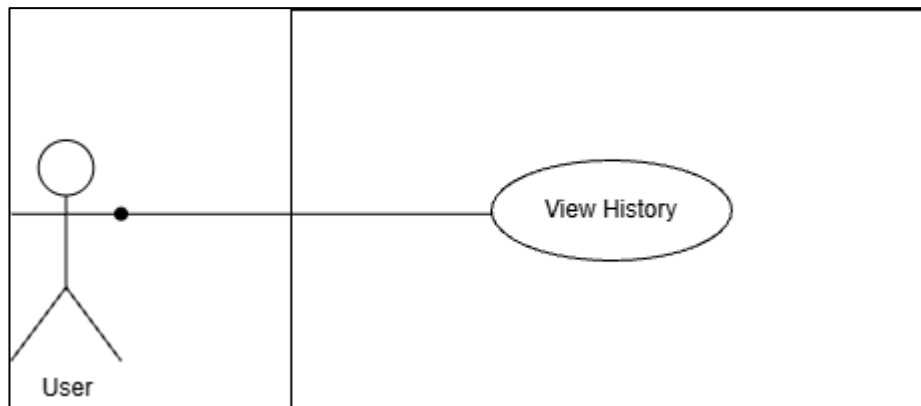


Figure 3.5: Use Case Diagram for View History

Table 3.12: Use Case for View History

Field	Details
Actors	User
Description	User views past text analyses.
Pre-Condition	User must be logged in.
Steps	1. User navigates to history page. 2. System retrieves all user's past analyses. 3. System displays list with dates, titles, and metrics. 4. User can filter or search history. 5. User can select an analysis to view details.
Post-Condition	Analysis history is displayed.

3.2.6 Use Case #5: Edit Profile

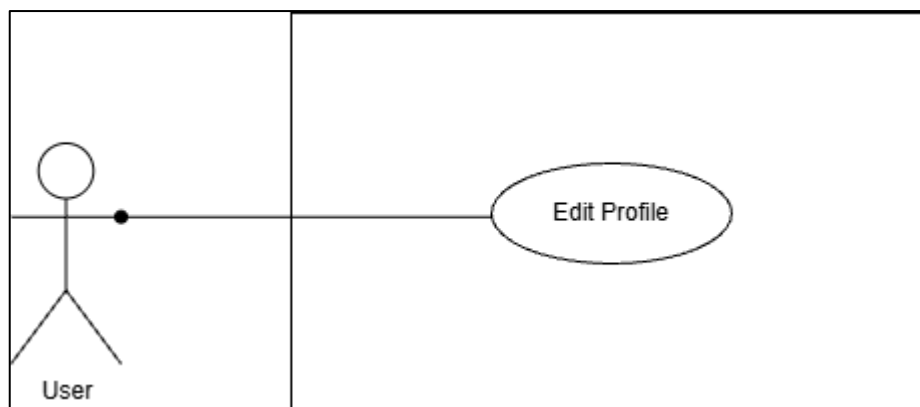


Figure 3.6: Use Case Diagram for Edit Profile

Table 3.13: Use Case for Edit Profile

Field	Details
Actors	User
Description	User updates their profile information.
Pre-Condition	User must be logged in.
Steps	1. User navigates to profile page. 2. User modifies name, email, profile picture, or password. 3. User saves changes. 4. System validates and updates information.
Post-Condition	User profile is updated.

3.2.7 Use Case #6: View Dashboard

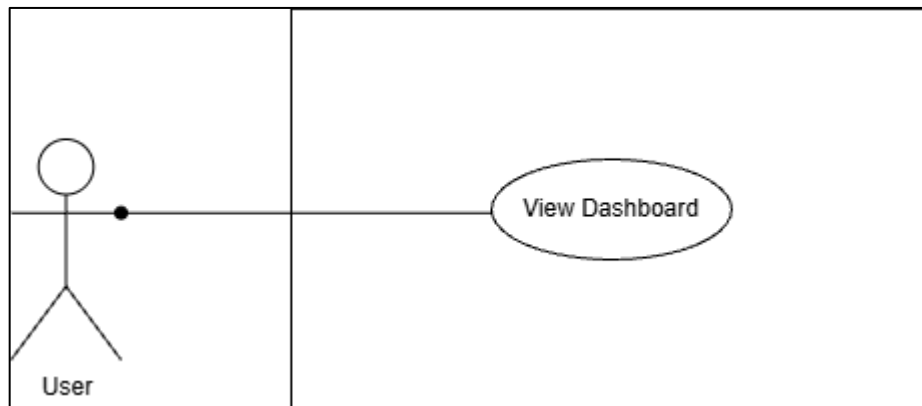


Figure 3.7: Use Case Diagram for View Dashboard

Table 3.14: Use Case for View Dashboard

Field	Details
Actors	User
Description	User views their main dashboard with recent analyses and statistics.
Pre-Condition	User must be logged in.
Steps	1. User navigates to dashboard. 2. System retrieves user's recent analyses. 3. System displays recent analysis history and quick stats. 4. System shows available actions.
Post-Condition	Dashboard is displayed with user's data and quick actions.

3.2.8 Use Case #7: Analyze Text Readability

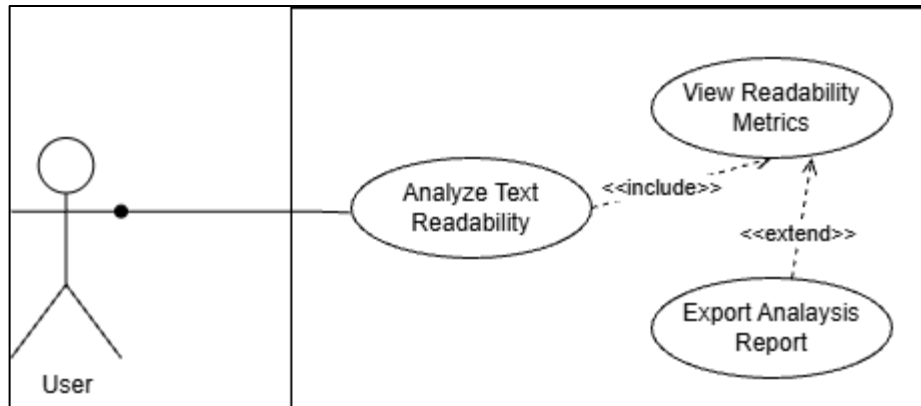


Figure 3.8: Use Case Diagram for Analyze Text Readability

Table 3.15: Use Case for Analyze Text Readability

Field	Details
Actors	User
Description	User submits text for readability analysis.
Pre-Condition	User must be logged in.
Steps	1. User submits text (via upload, paste, or dictation). 2. System processes text with NLP engine. 3. System calculates and displays readability metrics. 4. System identifies difficult words and complex sentences. 5. System displays results. 6. User may optionally export analysis report.
Post-Condition	Text analysis is saved and results are displayed.
Relationships	Includes: View Readability Metrics Extends: Export Analysis Report

3.2.9 Use Case #8: Choose Rewrite Level

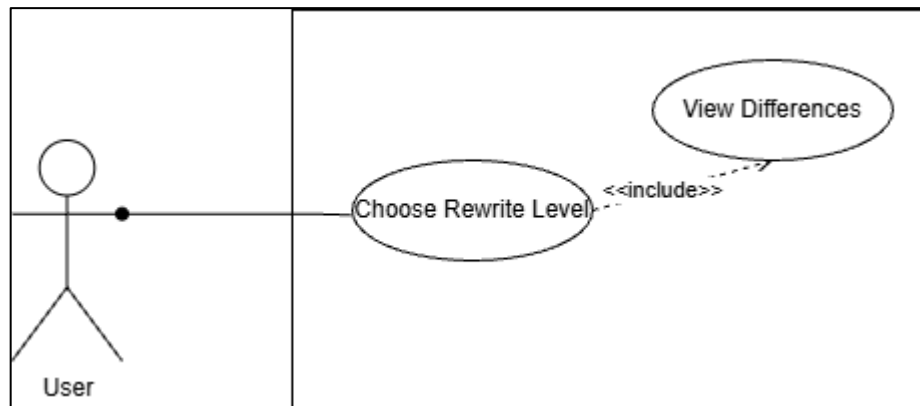


Figure 3.9: Use Case Diagram for Choose Rewrite Level

Table 3.16: Use Case for Choose Rewrite Level

Field	Details
Actors	User
Description	User selects target grade level for text simplification.
Pre-Condition	User must have analyzed text with readability metrics.
Steps	1. User views current text grade level. 2. User selects target grade level (e.g., 6th, 8th, 10th grade). 3. System sends text to rewrite engine for simplification. 4. System generates simplified version. 5. System displays comparison view with differences highlighted.
Post-Condition	Simplified text is generated and displayed alongside original.
Relationships	Includes: View Differences

3.2.10 Use Case #9: View All Analyses

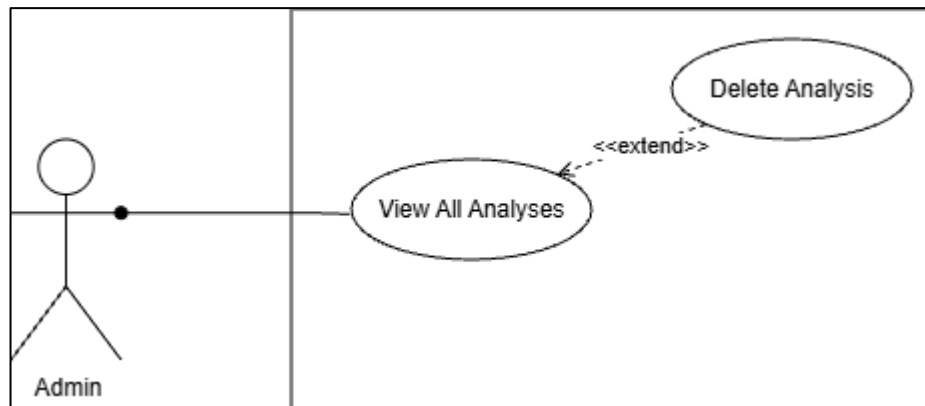


Figure 3.10: Use Case Diagram for View All Analyses

Table 3.17: Use Case for View All Analyses

Field	Details
Actors	Admin
Description	Admin views all text analyses in the system.
Pre-Condition	Admin must be logged in.
Steps	1. Admin navigates to analyses page. 2. System retrieves all analyses from all users. 3. System displays list with user info, dates, and metrics. 4. Admin can filter, search, or sort analyses. 5. Admin may delete an analysis.
Post-Condition	All system analyses are displayed to admin.
Relationships	Extends: Delete Analysis

3.2.11 Use Case #10: View Usage Statistics

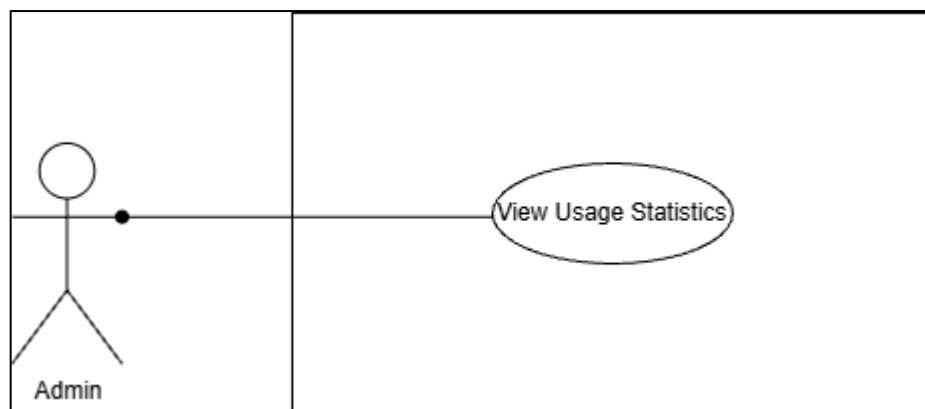


Figure 3.11: Use Case Diagram for View Usage Statistics

Table 3.18: Use Case for View Usage Statistics

Field	Details
Actors	Admin
Description	Admin views system-wide usage statistics and metrics.
Pre-Condition	Admin must be logged in.
Steps	1. Admin navigates to statistics page. 2. System calculates total users, analyses, and activity. 3. System displays average metrics and usage statistics.
Post-Condition	System usage statistics are displayed to admin.

3.2.12 Use Case #11: Manage User Accounts

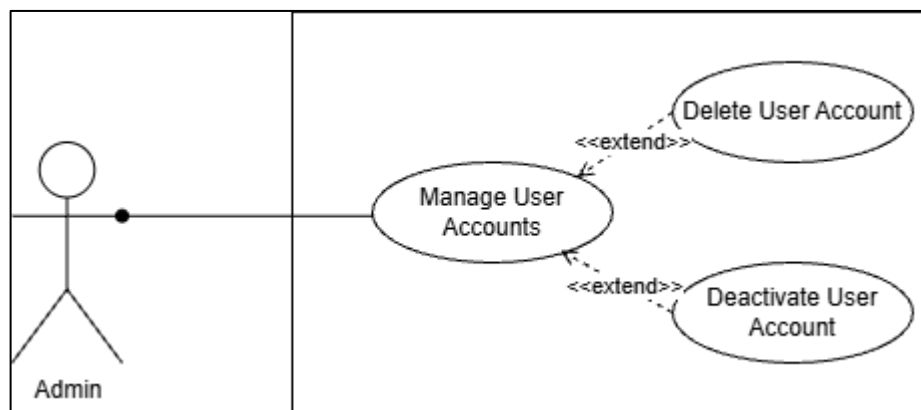


Figure 3.12: Use Case Diagram for Manage User Accounts

Table 3.19: Use Case for Manage User Accounts

Field	Details
Actors	Admin
Description	Admin manages user accounts in the system.
Pre-Condition	Admin must be logged in.
Steps	1. Admin navigates to user management page. 2. System displays list of all users. 3. Admin can view user details and activity. 4. Admin can modify user permissions or status. 5. Admin may delete or deactivate user accounts.
Post-Condition	Admin can view and manage all user accounts.
Relationships	Extends: Delete User Account, Deactivate User Account

3.2.13 Use Case #12: Logout

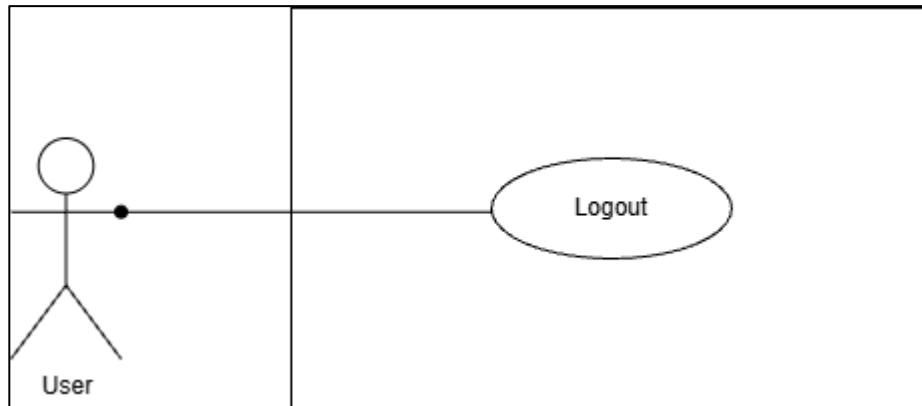


Figure 3.13: Use Case Diagram for Logout

Table 3.20: Use Case for Logout

Field	Details
Actors	User
Description	User ends their current session.
Pre-Condition	User must be logged in.
Steps	1. User clicks logout button. 2. System terminates session. 3. System clears authentication tokens.
Post-Condition	User is logged out and redirected to login page.

3.3 External Interface Requirements

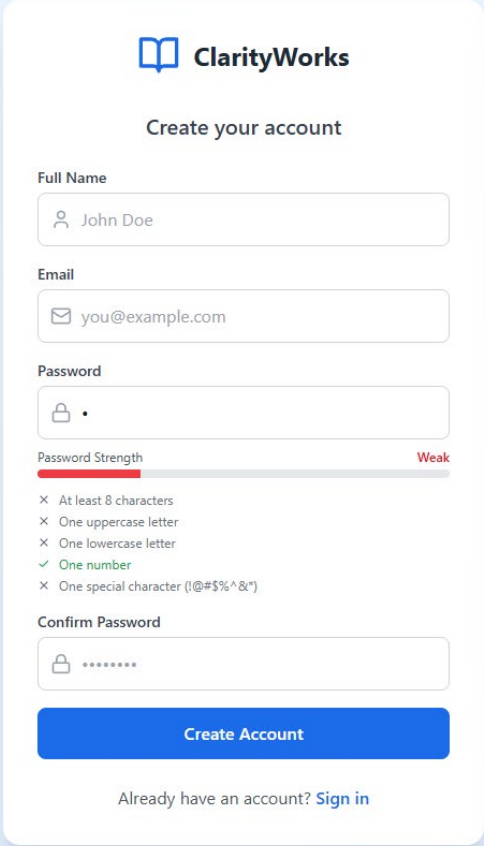
3.3.1 User Interfaces

The ClarityWorks platform provides a modern, responsive, and intuitive user interface tailored for two primary user roles: Users and Administrators. Each role is presented with role-specific screens and functionalities to ensure a streamlined and accessible experience.

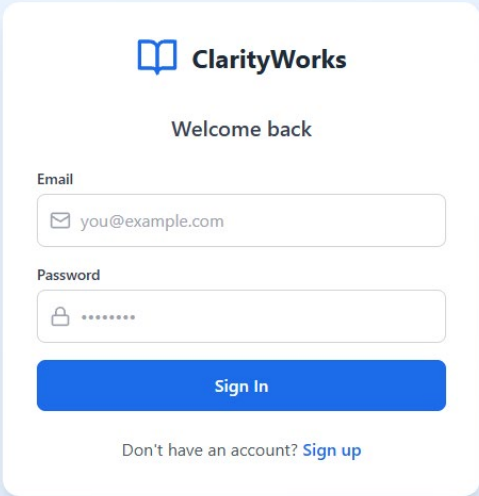
3.3.1.1 Regular User Interfaces

• Login & Registration Screens

Secure forms for account registration, login, and password management. Common UI actions include "Sign In," "Create Account," and "Sign up." Registration includes real-time password strength validation with checklist (8+ characters, uppercase, lowercase, number, special character). Inline validation messages display for incorrect credentials or missing inputs.



The registration form for ClarityWorks is displayed on a light blue background. It features a white card with the ClarityWorks logo at the top. Below the logo, the text "Create your account" is centered. The form includes three input fields: "Full Name" with the placeholder "John Doe", "Email" with the placeholder "you@example.com", and "Password" with a lock icon and a dot. Below the password field, a "Password Strength" indicator shows a red bar and the word "Weak". A list of requirements follows: "At least 8 characters", "One uppercase letter", "One lowercase letter", "One number" (checked), and "One special character (!@#%&*)". A "Confirm Password" field with a lock icon and dots is below. A blue "Create Account" button is at the bottom, followed by the text "Already have an account? [Sign in](#)".

Figure 3.14: Registration

The login form for ClarityWorks is displayed on a light blue background. It features a white card with the ClarityWorks logo at the top. Below the logo, the text "Welcome back" is centered. The form includes two input fields: "Email" with the placeholder "you@example.com" and "Password" with a lock icon and dots. A blue "Sign In" button is at the bottom, followed by the text "Don't have an account? [Sign up](#)".

Figure 3.15: Login

• User Dashboard

Personalized landing page showing welcome message, quick action cards ("Start New Analysis," "View History"), statistics overview (Total Analyses, Avg. Flesch Score, Avg. Grade Level, Words Analyzed), and recent analyses list with titles, dates, and grade level badges.

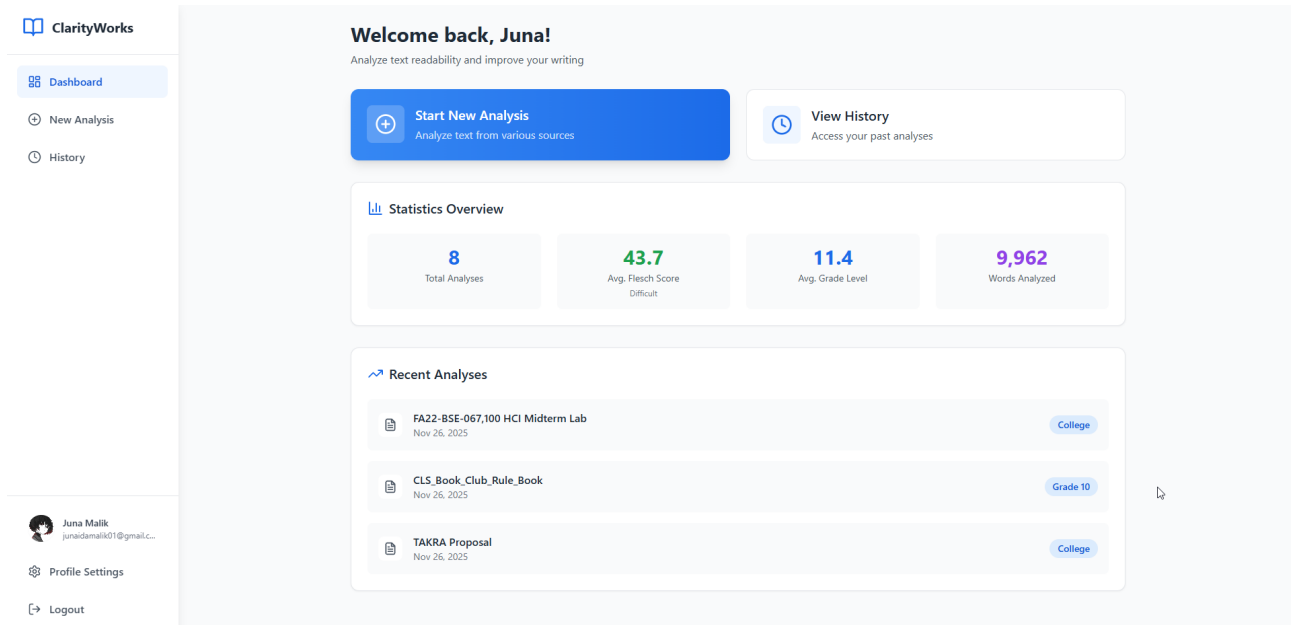


Figure 3.16: User Dashboard

• New Analysis Screen

Multi-tab interface for text input: Text (paste), PDF, DOC/DOCX, Image (OCR), Voice. Includes optional title field, drag-and-drop upload area with file size limits, extracted text preview with character/word count, and "Analyze Text" button.

ClarityWorks

Dashboard

New Analysis

History

Juna Malik
junaidemalik01@gmail...

Profile Settings

Logout

New Analysis

Enter or upload text to analyze its readability

Title (optional)
Give your analysis a title...

Text

PDF

DOC/DOCX

Image (OCR)

Voice

Click to upload or drag and drop

PDF files up to 10MB

Analyze Text

Figure 3.17: New Analysis

Title (optional)
FA22-BSE-067 SCD A2

Text

PDF

DOC/DOCX

Image (OCR)

Voice

✓ Extracted text from 13 pages

Click to upload or drag and drop

PDF files up to 10MB

Extracted Text (edit if needed)

Case Study – Addressing Complex Software Design and Development Challenges
By:
Junaid Ahsan Malik
FA22-BSE-067
Submitted to:
Ms. Maira Afzal

2515 words19635 characters

Analyze Text

Figure 3.18: Upload Options

• Analysis Results Screen

Displays analysis title, timestamp, and "Export PDF" button. Shows key metrics cards (Grade Level with complexity badge, Flesch Ease, Word Count, Difficult Words). Includes visualizations: Readability Scores radar chart, Grade Level gauge, Text Statistics bar chart, Word Difficulty pie chart, Top Difficult Words bar chart, and detailed metrics table.

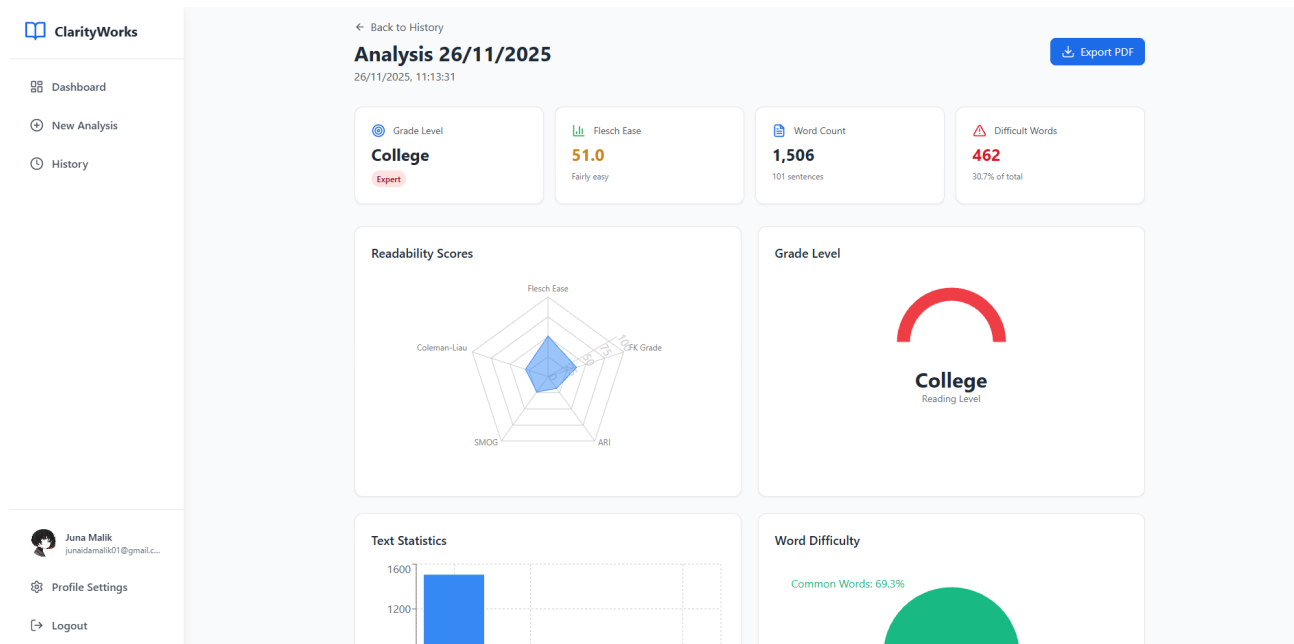


Figure 3.19: Analysis Results

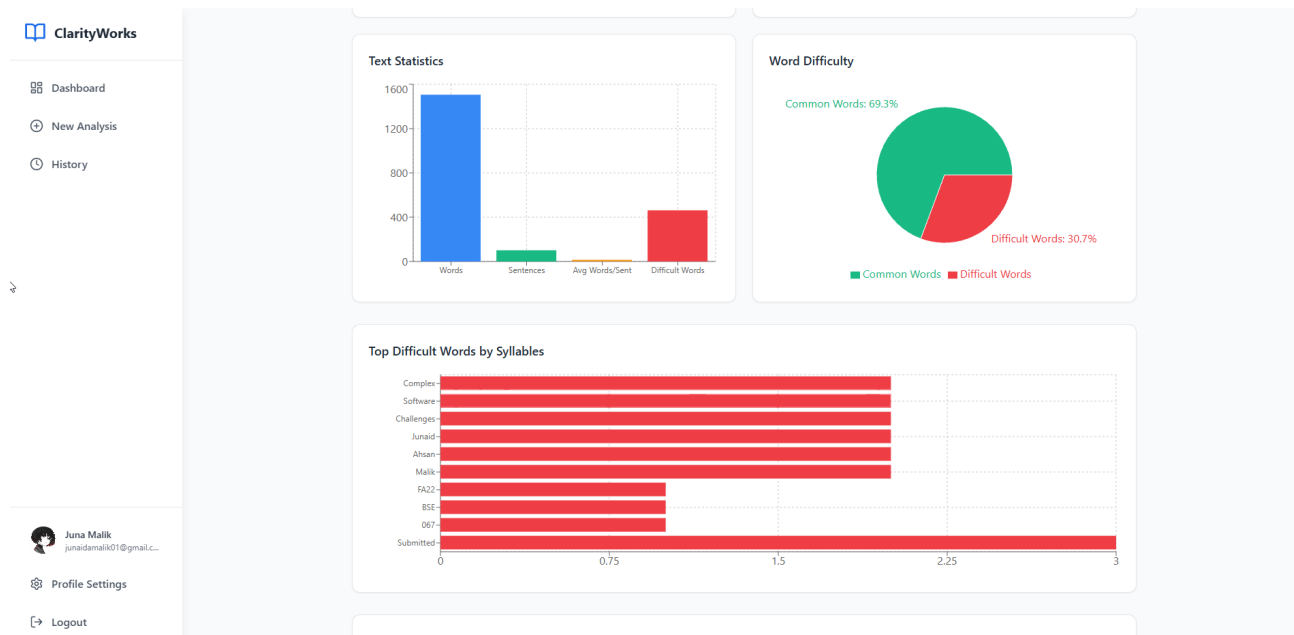


Figure 3.20: Difficult Words Ranking

• Detailed Analysis View

Shows difficult words and complex sentences highlighted with colored backgrounds. Each flagged item displays reason (e.g., "Long sentence (34 words)"), Flesch Score, and word count.

Each country will have different rules and regulations, their own constraints that come with the country and its culture, culture and lifestyle differences meaning that customers have their own prefer...

Reason: Long sentence (34 words) Flesch Score: 40.4 Words: 34

Like this, we can see that we won't ever really have a software that we can say is 'done' with its requirements; the requirements will consistently be evolving and changing during the design and devel...

Reason: Long sentence (41 words) Flesch Score: 41.4 Words: 41

Conflicting priorities arise from our system having so many different departments

Reason: Complex structure (Flesch: 26.5) Flesch Score: 26.5 Words: 11

Such a complex system will have different departments that need to do different tasks, but sometimes their priorities will overlap and we will have to manage this

Reason: Long sentence (27 words) Flesch Score: 51 Words: 27

For example, the procurement department wishes for a flexible design so they can adapt to any changes

Reason: Contains 4 complex words Flesch Score: 55.2 Words: 17

Meanwhile, manufacturing needs a rigid, solid design so they can incorporate all their automation requirements into the project

Reason: Complex structure (Flesch: 28.8) Flesch Score: 28.8 Words: 18

Figure 3.21: Difficult Sentences

• Analysis History Screen

Searchable and filterable table listing past analyses. Includes search bar, grade level filter dropdown, and table columns: Title, Date, Words, Flesch Score (color-coded), Grade Level (badge), Actions (view/delete icons).

ClarityWorks

Dashboard

New Analysis

History

Juna Malik
junardamalik01@gmail.com

Analysis History

View and manage your past analyses

Search by title or content...

All Grade Levels

Title	Date	Words	Flesch Score	Grade Level	Actions
FA22-BSE-067,100 HCI Midterm Lab	Nov 26, 2025, 09:06 AM	3,149	37.6	College	View Delete
CLS_Book_Club_Rule_Book	Nov 26, 2025, 06:21 AM	695	57.3	Grade 10	View Delete
TAKRA Proposal	Nov 26, 2025, 06:21 AM	403	41.4	College	View Delete
MAD Questions	Nov 26, 2025, 06:13 AM	2,344	45.2	Grade 3	View Delete
CV Allauddin 0723 og	Nov 26, 2025, 06:12 AM	757	26.7	Grade 3	View Delete
PPIT Chapter 3	Nov 23, 2025, 10:36 AM	710	27.3	Grade 3	View Delete
CN Assignment 4	Nov 23, 2025, 10:30 AM	1,736	73.1	Grade 3	View Delete

Figure 3.22: Analysis History

- **Profile Settings Screen**

Tabbed interface with Profile tab (profile picture upload/remove, Full Name, Email, "Save Changes" button) and Password tab. Success/error messages provide feedback on actions.

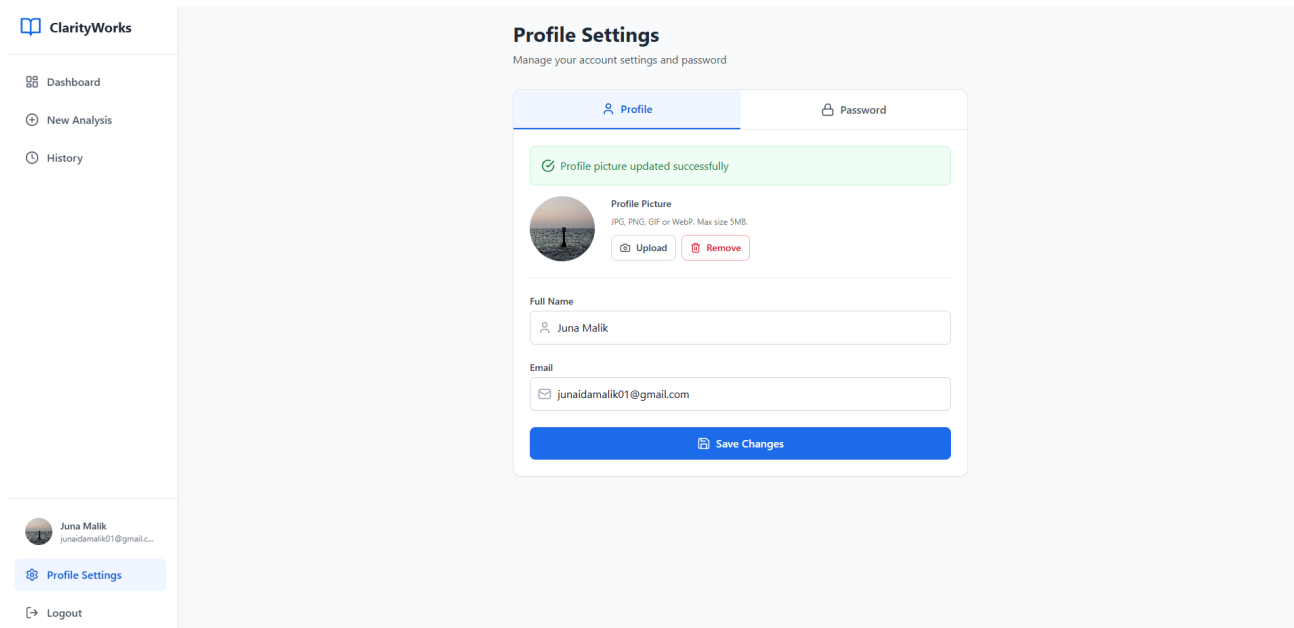


Figure 3.23: Edit Profile

- **Export Report (PDF)**

Generated report includes ClarityWorks branding, analysis date/timestamp, summary section (Grade Level, Complexity, Flesch Ease, Confidence), and detailed text statistics (Word Count, Sentence Count, Paragraph Count, Avg Words/Sentence, Avg Syllables/Word, Difficult Words).

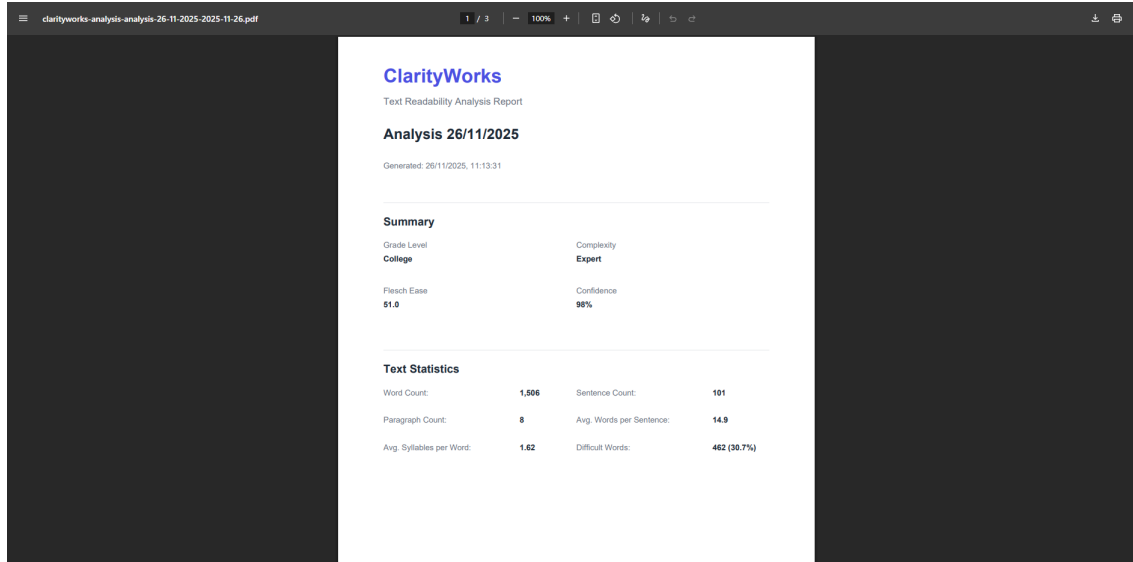


Figure 3.24: Analysis Report

3.3.1.2 Administrator Interfaces

• Admin Dashboard

High-level summary showing Total Users, Total Analyses, Avg Reading Ease, Words Analyzed with activity indicators (e.g., "+2 this week"). Includes Grade Level Distribution bar chart, User Status pie chart, Recent Activity list, and Top Users ranking.

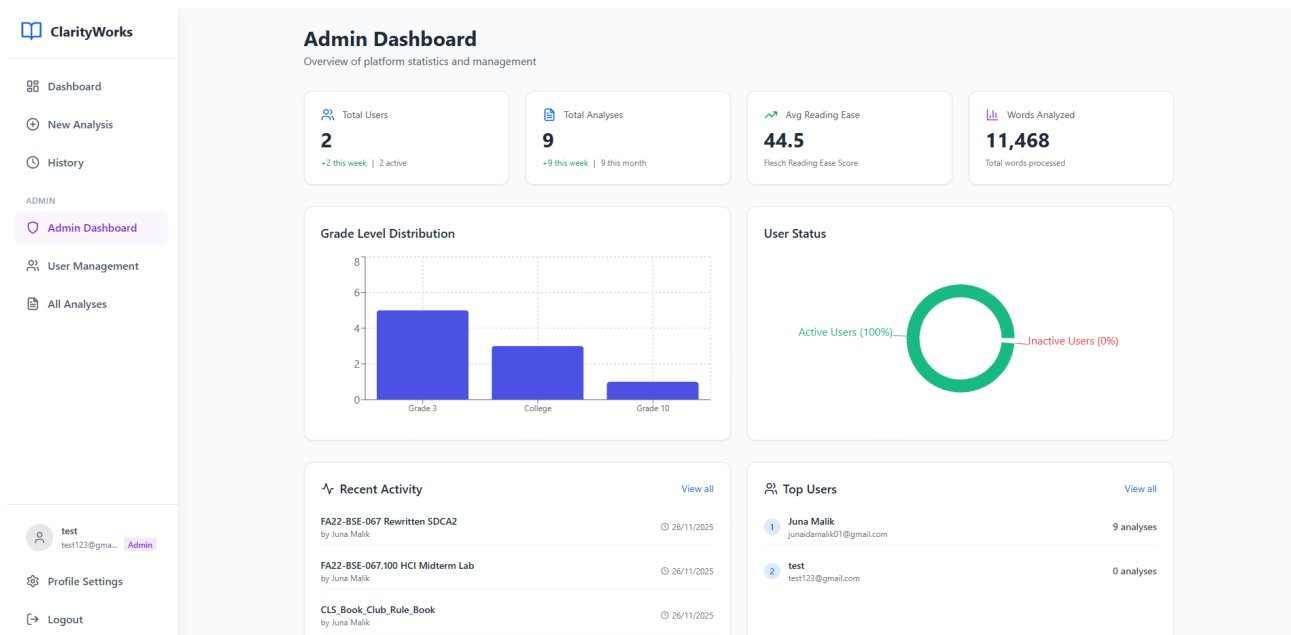


Figure 3.25: Admin Dashboard

• User Management Screen

Displays all registered users in searchable table with filters for roles and status. Shows User, Role (badge), Status (Active/Inactive badge), Analyses count, Joined date, and Actions (view details, activate/deactivate, delete icons).

USER	ROLE	STATUS	ANALYSES	JOINED	ACTIONS
test test123@gmail.com	Admin	Active	0	23/11/2025	View Details, Activate/Deactivate, Delete
Juna Malik junaidamalik01@gmail.com	User	Active	9	22/11/2025	View Details, Activate/Deactivate, Delete

Figure 3.26: User Management

• Analysis Management Screen

System-wide view of all analyses with search functionality. Table columns: Title, User (name/email), Grade Level, Complexity (badge), Words, Flesch Score, Created date, Actions (view/delete icons).

TITLE	USER	GRADE LEVEL	COMPLEXITY	WORDS	FLESCH SCORE	CREATED	ACTIONS
FA22-BSE-067 Rewritten SDCA2	Juna Malik junadmalik01@gmail.com	College	Expert	1,506	51.0	26/11/2025	View Delete
FA22-BSE-067.100 HCI Midterm Lab	Juna Malik junadmalik01@gmail.com	College	Expert	3,149	37.6	26/11/2025	View Delete
CLS_Book_Club_Rule_Book	Juna Malik junadmalik01@gmail.com	Grade 10	Advanced	695	57.3	26/11/2025	View Delete
TAKRA Proposal	Juna Malik junadmalik01@gmail.com	College	Expert	403	41.4	26/11/2025	View Delete
MAD Questions	Juna Malik junadmalik01@gmail.com	Grade 3	Elementary	2,344	45.2	26/11/2025	View Delete
CV Allaudin 0723 og	Juna Malik junadmalik01@gmail.com	Grade 3	Elementary	757	26.7	26/11/2025	View Delete
PPIT Chapter 3	Juna Malik junadmalik01@gmail.com	Grade 3	Elementary	710	27.3	23/11/2025	View Delete
CN Assignment 4	Juna Malik junadmalik01@gmail.com	Grade 3	Elementary	1,736	73.1	23/11/2025	View Delete
Inter-Society Capture Clash 2025 — Submission ...	Juna Malik junadmalik01@gmail.com	Grade 3	Elementary	168	40.8	22/11/2025	View Delete

Figure 3.27: Analysis Management

Common UI Elements and Standards

• Navigation Bar / Sidebar

Present on all screens with ClarityWorks logo, user profile widget at bottom, and role-specific menu items (Dashboard, New Analysis, History for users; Admin Dashboard, User Management, All Analyses for admins).

• Standard Buttons

Actions like "Sign In," "Create Account," "Analyze Text," "Save Changes," "Export PDF" are consistently styled with primary blue color. Icon buttons for view/delete/edit use consistent iconography.

• Feedback Messages

Success messages displayed in green banner (e.g., "Profile picture updated successfully"). Error messages use inline red text. Real-time validation on forms.

• Responsive Design

All screens optimized for desktops, laptops, and tablets with adaptive layouts.

• Consistent Layouts

Structured use of cards, tables, charts, badges, and modals ensures clean data presentation across all modules.

3.3.2 Hardware Interfaces

The ClarityWorks platform will be a web-based application and will not require any direct integration with specialized hardware components. The system is expected to operate smoothly across standard computing devices by utilizing modern web browsers for all user interactions.

3.3.3 Software Interfaces

ClarityWorks integrates with various software components to deliver its functionality. This section details the connections between ClarityWorks and other software elements, including operating systems, databases, libraries, tools, and external services.

Operating System Interfaces

Windows/macOS

- The web application interfaces with modern browsers (Chrome, Firefox, Safari, Edge).
- Interfaces used:
 - Web Speech API (for voice dictation functionality)
 - File System Access API (for document uploads)

Database and Backend Services

PostgreSQL Database

- Used as the primary relational database.
- Stores data related to users, text analyses, readability metrics, simplified texts, and system statistics.
- Supports efficient querying and indexing for fast retrieval of analysis history.

Flask/FastAPI Backend

- Backend framework used to host REST API endpoints.
- Handles authentication, text processing, simplification, and admin functions.
- Provides fast, asynchronous HTTP endpoints for the React frontend.

Backend NLP & Text Processing Services

spaCy (Python NLP Library)

- Used for core text processing: tokenization, sentence segmentation, POS tagging.

NLTK (Natural Language Toolkit)

- Supplementary library for syllable counting and linguistic features.
- Used in readability metric calculations.

Text Simplification Engine

- Rule-based simplification with:
 - Word replacement dictionaries
 - Sentence splitting algorithms
 - Structure simplification rules

Gemini API

- External AI service for enhanced text simplification suggestions and NLG.
- Interface: HTTPS REST API with JSON requests/responses.

Document Processing Libraries**PyPDF2**

- Extracts text from uploaded PDF documents.

python-docx

- Extracts text from .docx files.
- Generates .docx export reports with formatting.

Tesseract OCR (via pytesseract)

- Extracts text from uploaded images (.jpg, .png).
- Open-source OCR engine for image-to-text conversion.

3.3.4 Communications Interfaces

The ClarityWorks system relies on secure, web-based communication between the React frontend application, Flask/FastAPI backend services, and the PostgreSQL database. All communication is performed over HTTPS, ensuring encrypted, secure data transfer to protect user credentials, text submissions, analysis results, and profile data. The system uses standard REST APIs with JSON-formatted messages for requests and responses, providing a consistent and interoperable communication model across all software components. The Flask/FastAPI backend exposes RESTful API endpoints for authentication, text analysis, simplification,

history management, and admin functions. The React frontend communicates with these endpoints through authenticated HTTPS calls using JWT tokens or session cookies.

4 Other Non-functional Requirements

4.1 Performance Requirements

The following performance requirements are proposed for the ClarityWorks platform:

- **Transaction Response Time:**

All user-initiated transactions such as login, registration, text submission, or profile updates should be processed within 3 seconds under normal load. Under high server load, transactions must still complete within 5 seconds.

- **Analysis Processing Time:**

Text readability analysis should complete within 5 seconds for texts up to 5,000 words. Texts between 5,000-10,000 words should complete within 10 seconds.

- **Simplification Processing Time:**

Text simplification should complete within 10 seconds for texts up to 5,000 words, including API calls to external services if used.

- **Concurrent Users:**

The system should support a minimum of 50 concurrent users performing operations like analyzing text, viewing history, and managing profiles without significant performance degradation.

- **Page Load Time:**

The dashboard and key functional screens (New Analysis, History, Analysis Results) should load within 3 seconds on a standard broadband connection.

- **Scalability:**

The architecture will support horizontal scalability, allowing the application to run across multiple servers to accommodate increased user demand.

- **Data Consistency:**

Analysis results and user data updates should be reflected immediately across all relevant user interfaces without requiring page refresh.

4.2 Safety and Security Requirements

ClarityWorks is designed with a strong focus on security and data protection. The proposed system will adhere to modern web security standards to protect user data and ensure secure access.

- **User Authentication:**

All users must log in via a secure login system using hashed passwords (bcrypt).

- **Data Encryption:**

All communication between client and server will occur over HTTPS to ensure encrypted transmission. Sensitive data in the database (passwords) will be hashed before storage.

- **Access Control:**

Role-based access control (RBAC) will ensure that only authorized users (User, Admin) can access role-specific functionalities. Admins have elevated privileges for user and analysis management.

- **Data Privacy:**

All personal data (user profiles, analysis texts, history) will be securely stored and not shared with third parties without user consent.

- **Input Validation:**

All inputs will be validated on both client and server sides to mitigate threats like SQL injection, XSS, CSRF, and other common web vulnerabilities.

- **Session Management:**

Sessions will be securely managed with automatic expiration after 30 minutes of inactivity to prevent unauthorized access. Session tokens will be secure.

- **Rate Limiting:**

Rate limiting (max 100 analyses per day per user) will prevent abuse and ensure fair resource allocation.

- **Backup and Recovery:**

Regular automated database backups are recommended to minimize data loss risk during system failures.

- **No Direct Hardware Control:**

The web-based platform does not interact with hardware in a way that can result in physical damage or safety risks.

- **Compliance:**

The system is intended to comply with standard web privacy and data protection regulations including GDPR for European users.

Major Security Requirements:

- All passwords must be stored hashed (never in plain text).
- All sensitive data must be transmitted over HTTPS.
- Only authenticated users can access protected resources.
- Administrators can deactivate or remove user accounts upon detecting suspicious activity.
- All critical user actions (login attempts, analysis creation, profile changes) are logged for audit and security monitoring.

4.3 Software Quality Attributes

4.3.1 Reliability

- The system will handle unexpected errors gracefully using user-friendly error messages and comprehensive error logging.
- Core modules must maintain $\geq 95\%$ success rate for critical operations (login, text analysis, export) under normal load.
- Database transactions will use ACID properties to ensure data integrity.

4.3.2 Availability

- The system is proposed to be available 24/7, with planned maintenance periods communicated in advance and scheduled during low-traffic hours.
- Reliable hosting infrastructure (cloud platforms like AWS, Azure, or DigitalOcean) and regular data backups will ensure minimal downtime and high service continuity.
- Target uptime: 99.5% (allowing approximately 3.6 hours downtime per month).

4.3.3 Maintainability

- The codebase will follow Python/Flask and React best practices, utilizing MVC architecture for clear separation of concerns.
- All public functions and classes must have documentation coverage $\geq 80\%$.

- Code will be well-documented with inline comments, docstrings, and README files to facilitate future maintenance and upgrades.
- Version control (Git) will track all code changes with meaningful commit messages.

4.3.4 Portability

- The platform will be platform-independent, deployable on any server that supports Python 3.8+ and PostgreSQL 12+.
- All project dependencies will be managed using pip and requirements.txt for consistent environment setup.
- Docker containerization may be used for simplified deployment across different environments.

4.3.5 Usability

- The UI will be designed to be intuitive and responsive across all devices (desktop, tablet,) with consistent navigation and layouts.
- Consistent use of standard UI elements (buttons, forms, tables, charts) enhances user experience.
- Error and success messages will be clear, informative, and actionable.
- User flows will be optimized to minimize clicks and cognitive load (e.g., one-click access to key functions from dashboard).

4.3.6 Security

- All passwords must be stored hashed using bcrypt (never in plain text).
- The platform will undergo regular security audits and timely updates to address any discovered vulnerabilities.
- All critical user actions (login attempts, analysis creation, profile changes) must be logged for audit and security monitoring.

4.3.7 Design for Change (Flexibility)

The modular structure of the application will support easy addition of features such as:

- Additional readability formulas or linguistic metrics
- Advanced text simplification

- User collaboration features (sharing analyses, team workspaces)

4.3.8 Testability

- Core modules must support unit tests with $\geq 70\%$ code coverage.
- Backend APIs must provide test/sandbox endpoints for integration testing.
- Mock services must allow testing without real PostgreSQL or external APIs.

5 Design Description

5.1 Composite Viewpoint

5.1.1 Package Diagram (Logical)

The package diagram represents the modular structure of ClarityWorks. Shows interactions between key components such as Admin Panel, User Profile, Login/Signup, Text Transformation, Submit Text, Analysis History, and Export Files, each encapsulated for clarity and maintainability.

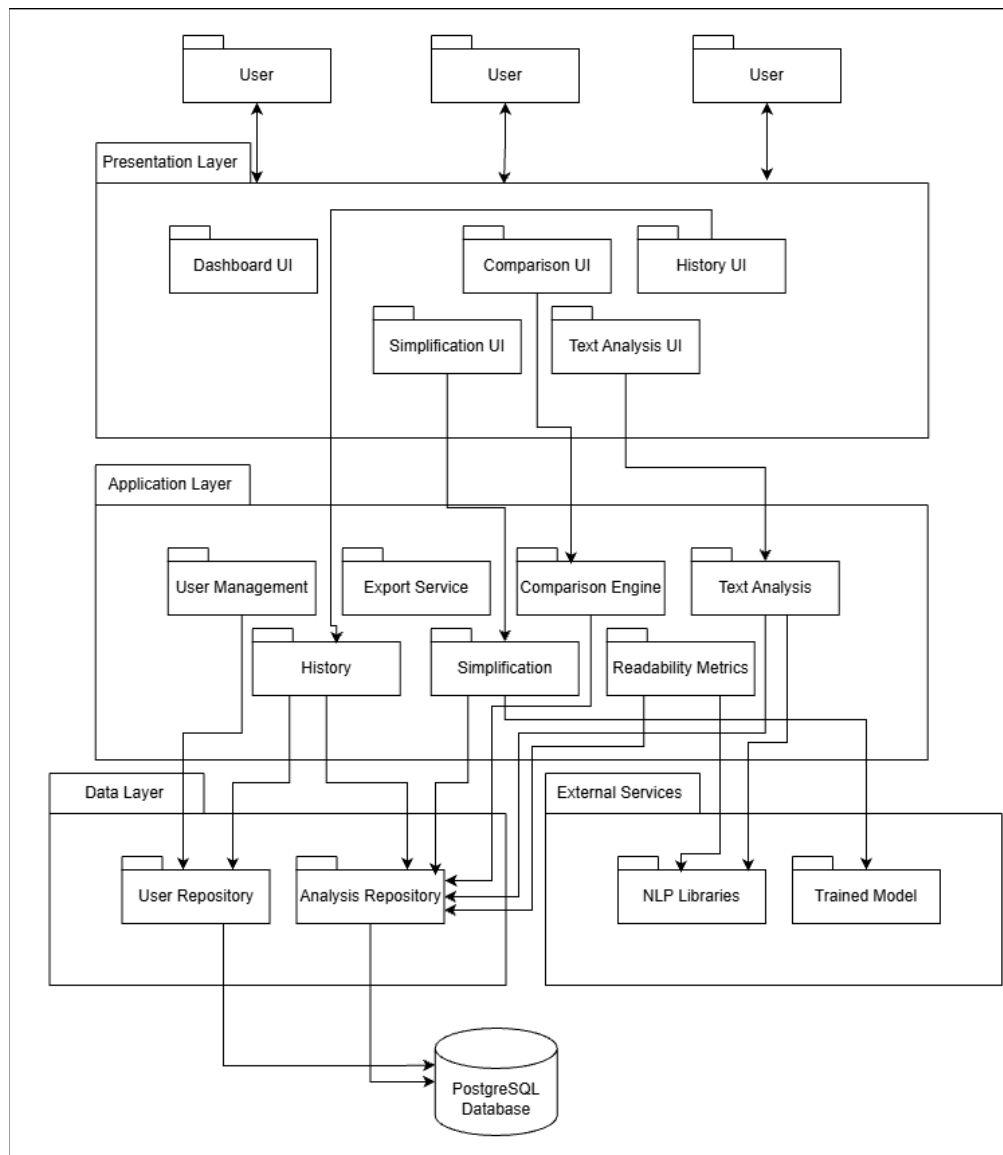


Figure 5.1: Package Diagram of the System

5.1.2 Deployment Diagram (Physical)

The deployment diagram illustrates how various components of ClarityWorks are deployed across the web server, database server, and external services. Highlights interactions with the PostgreSQL database, NLP libraries, and text processing services, ensuring smooth integration and operational efficiency.

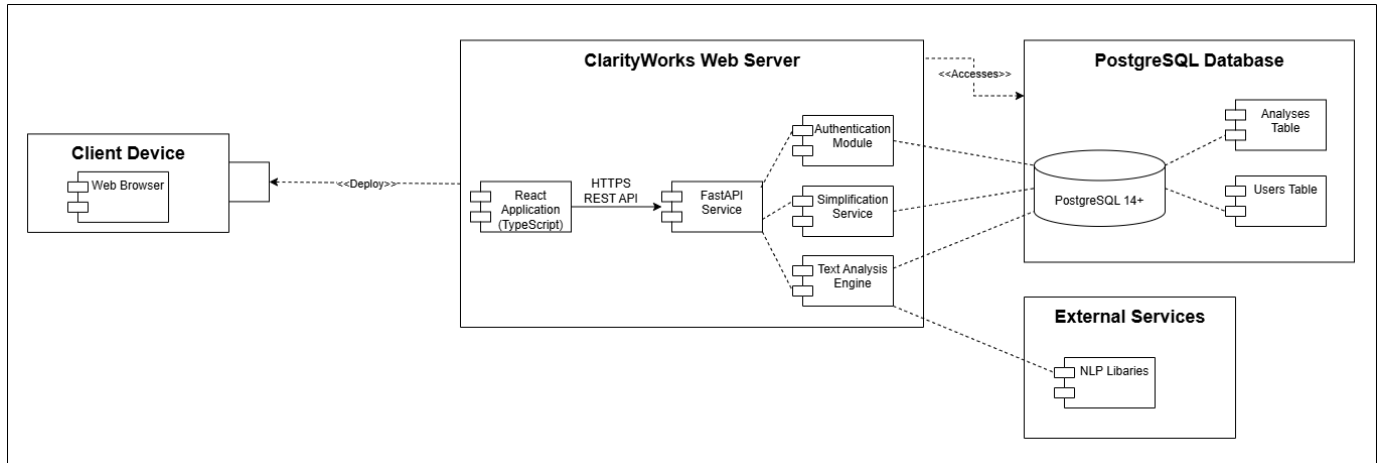


Figure 5.2: Deployment Diagram of the System

5.2 Logical Viewpoint

The logical viewpoint presents the structural design of ClarityWorks, focusing on the core building blocks of the system. The class diagram illustrates key entities such as User, RegularUser, Admin, TextAnalysis, ReadabilityMetrics, SimplifiedText, DifficultWord, and ComplexSentence. It defines their attributes, methods, and relationships, reflecting how different components interact to support text analysis, readability assessment, simplification, user roles, and system-wide operations.

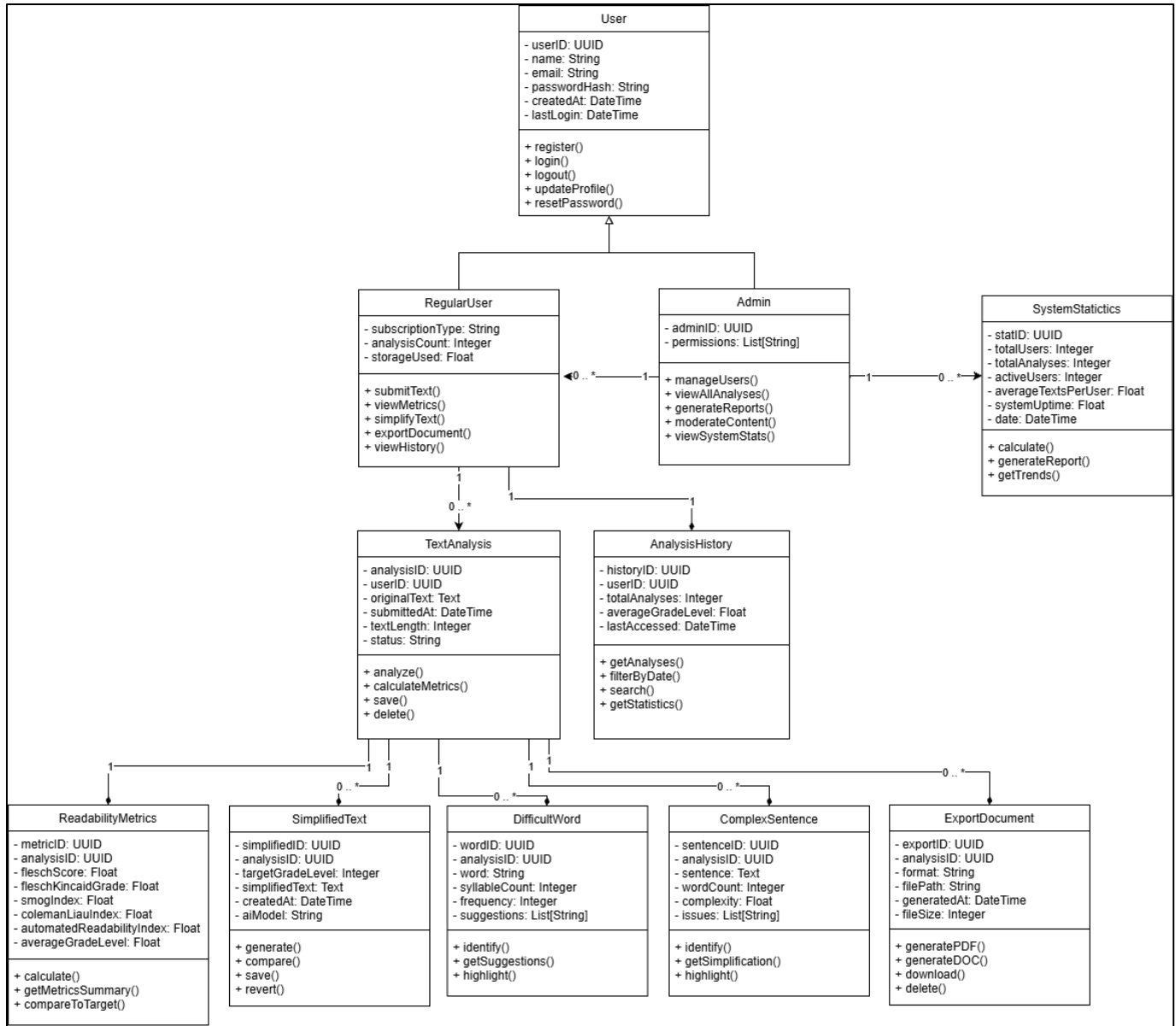


Figure 5.3: Class Diagram of the System

5.3 Information Viewpoint

The Entity-Relationship Diagram of ClarityWorks represents the data model of the system by outlining entities such as User and Analysis, along with their relationships. It provides a structured view of how data is organized and interlinked within the database.

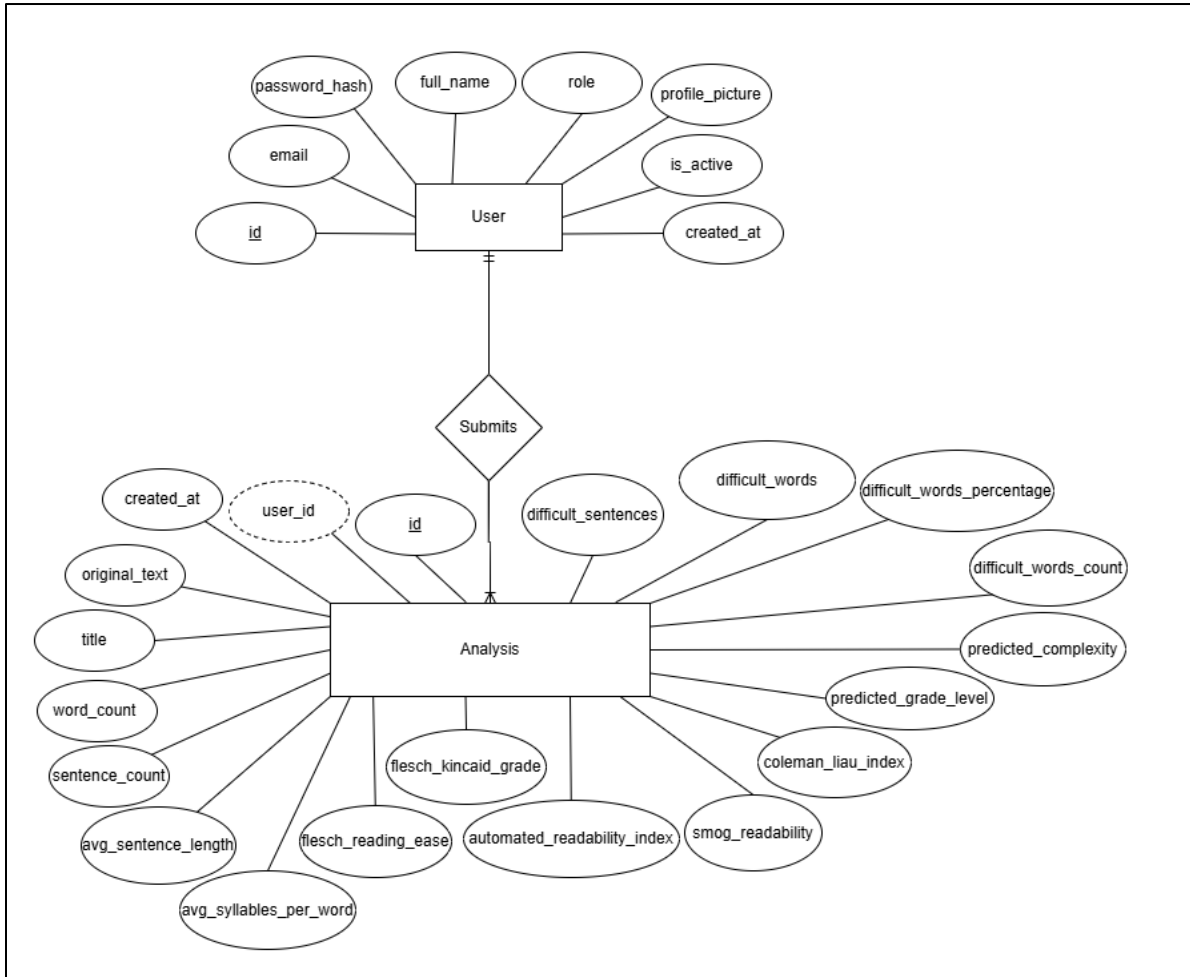


Figure 5.4: Entity Relationship Diagram of the System

5.4 Interaction Viewpoint

This viewpoint defines how system components and actors interact with one another across different use cases, such as user login, text analysis, simplification, and admin management. Sequence diagrams are used to illustrate the flow of messages between objects, ensuring clarity in task execution and system communication.

5.4.1 Sequence Diagram for Register

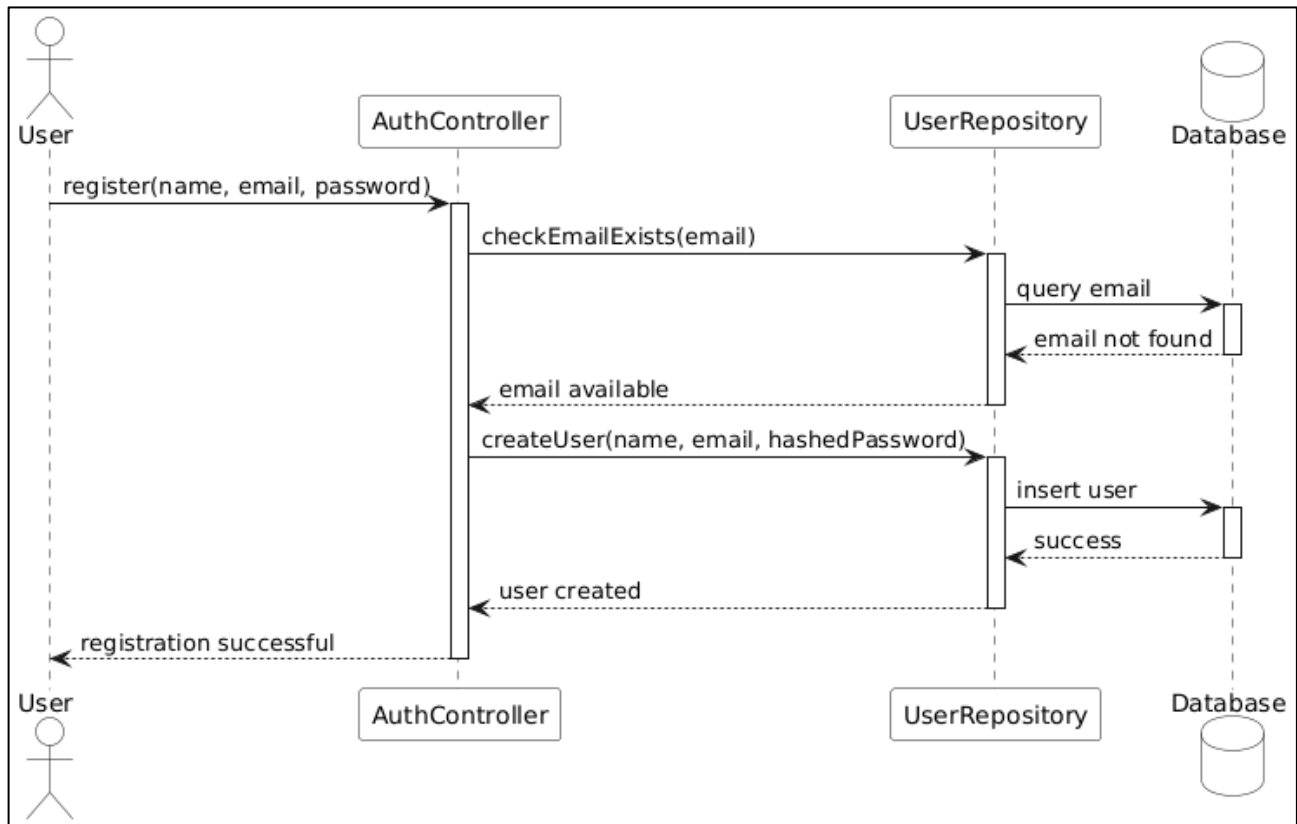


Figure 5.5: Sequence Diagram for Register

5.4.2 Sequence Diagram for Login

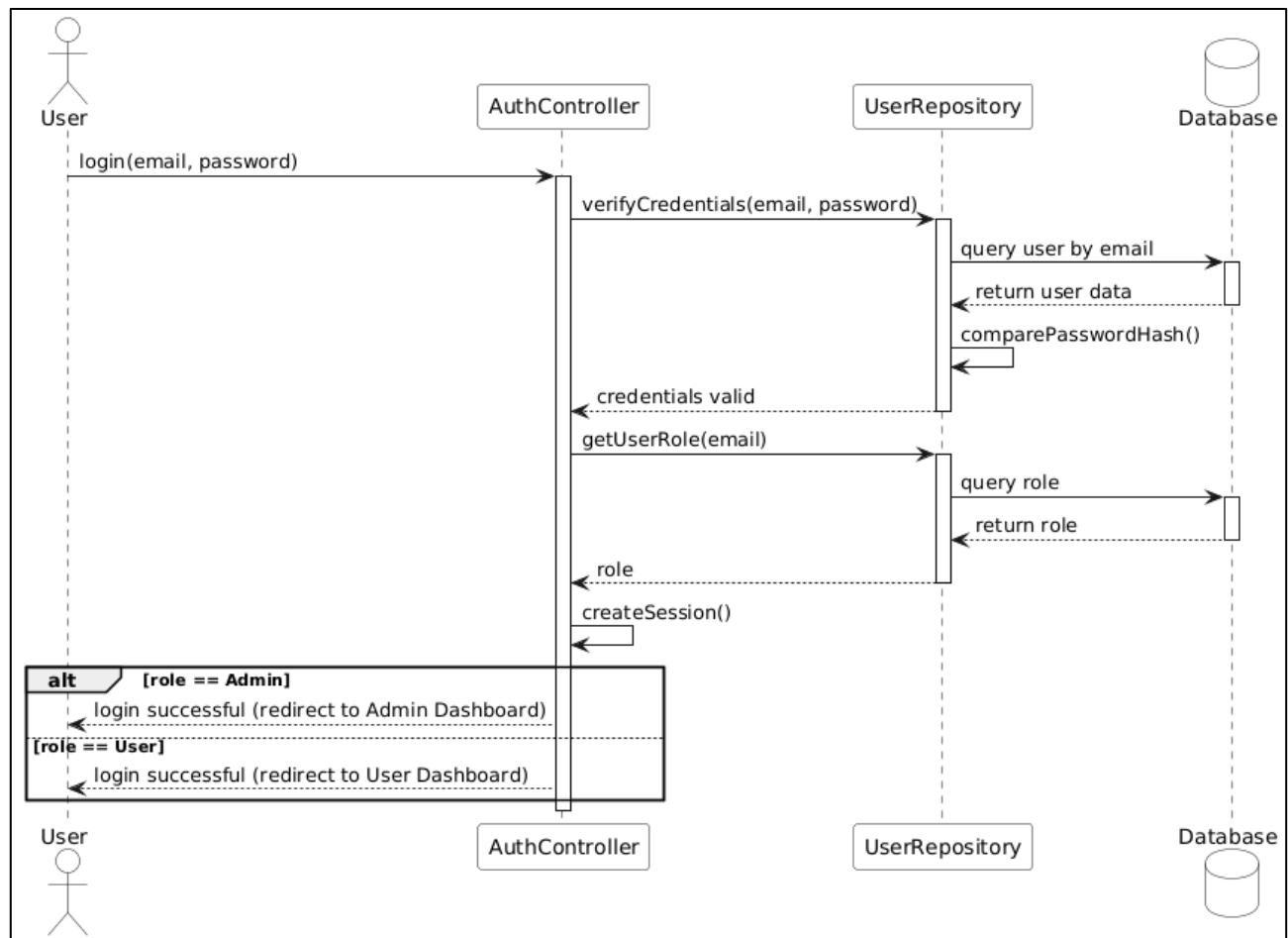


Figure 5.6: Sequence Diagram for Login

5.4.3 Sequence Diagram for Upload Text



Figure 5.7: Sequence Diagram for Upload Text

5.4.4 Sequence Diagram for View History

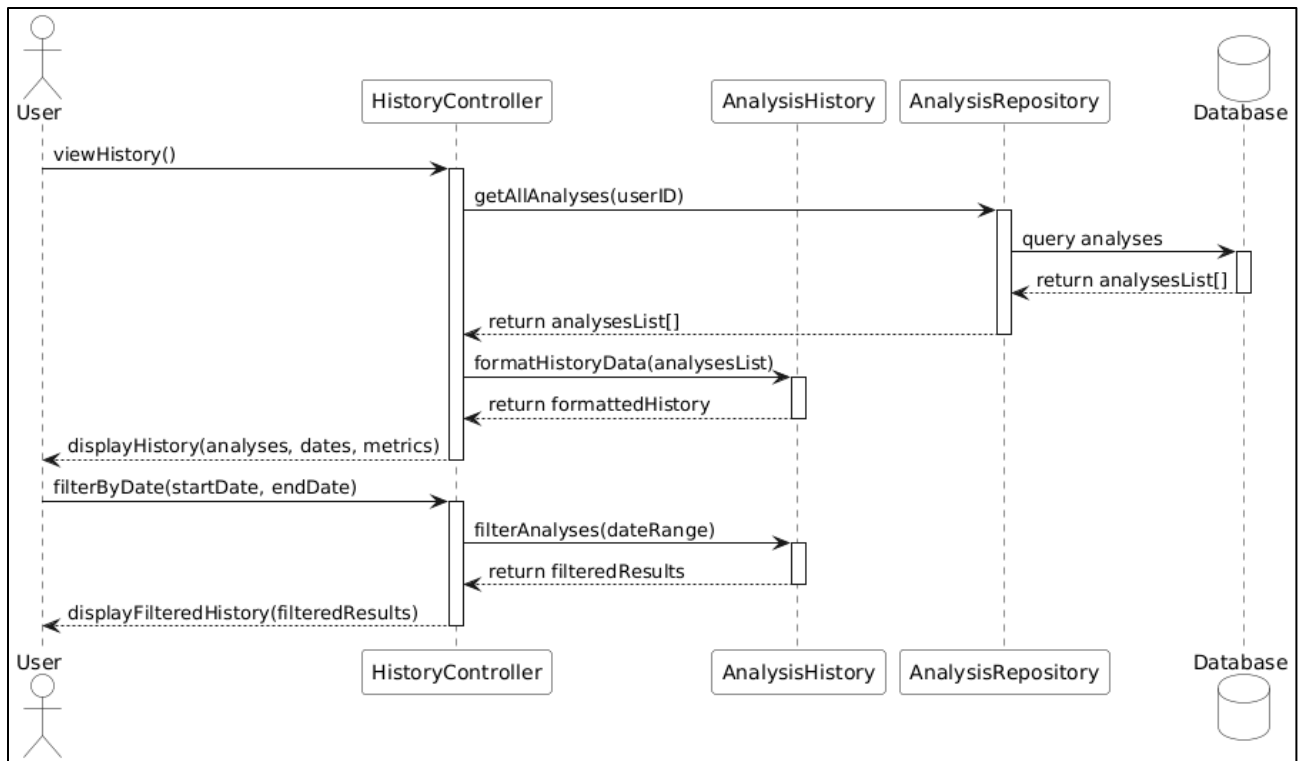


Figure 5.8: Sequence Diagram for View History

5.4.5 Sequence Diagram for Edit Profile

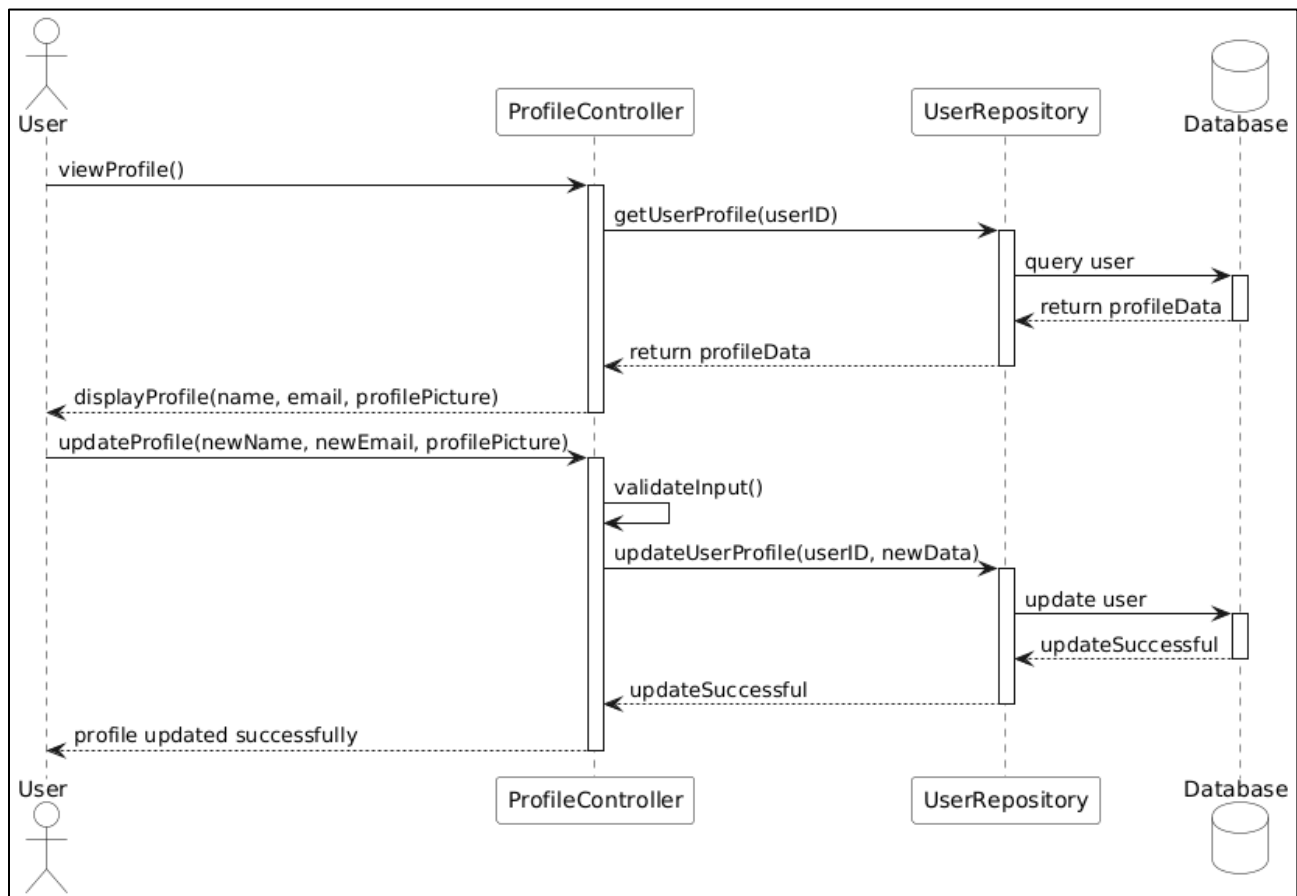


Figure 5.9: Sequence Diagram for Edit Profile

5.4.6 Sequence Diagram for View Dashboard

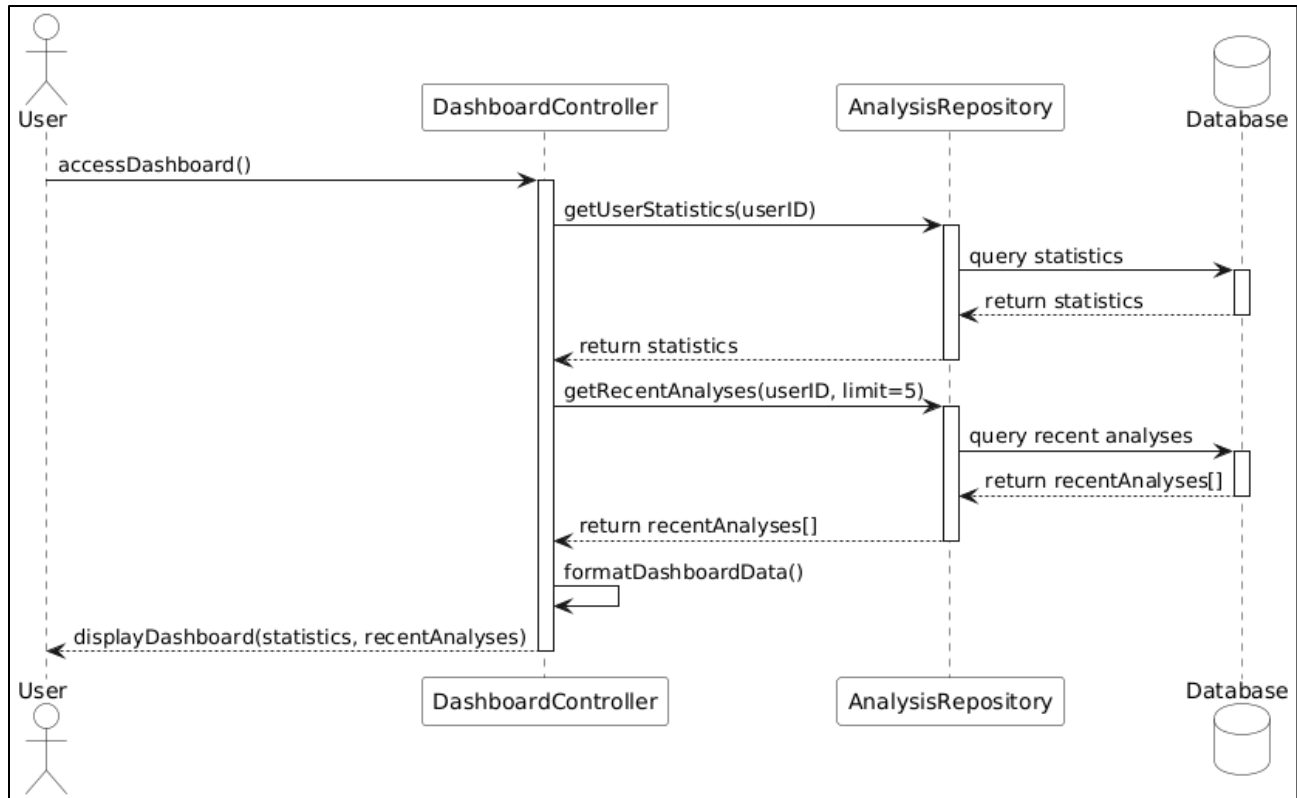


Figure 5.10: Sequence Diagram for View Dashboard

5.4.7 Sequence Diagram for Analyze Text Readability

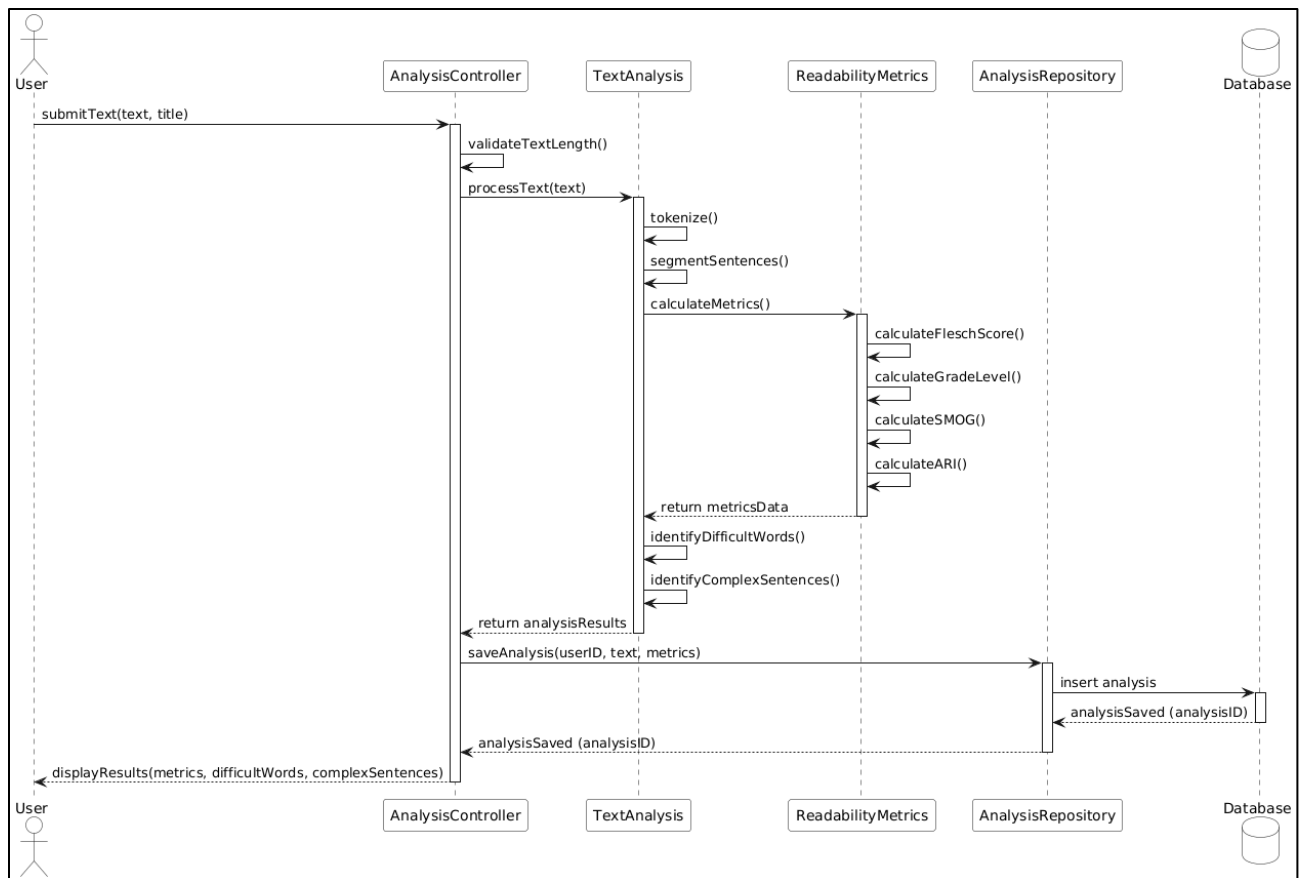


Figure 5.11: Sequence Diagram for Analyze Text Readability

5.4.8 Sequence Diagram for Choose Rewrite Level

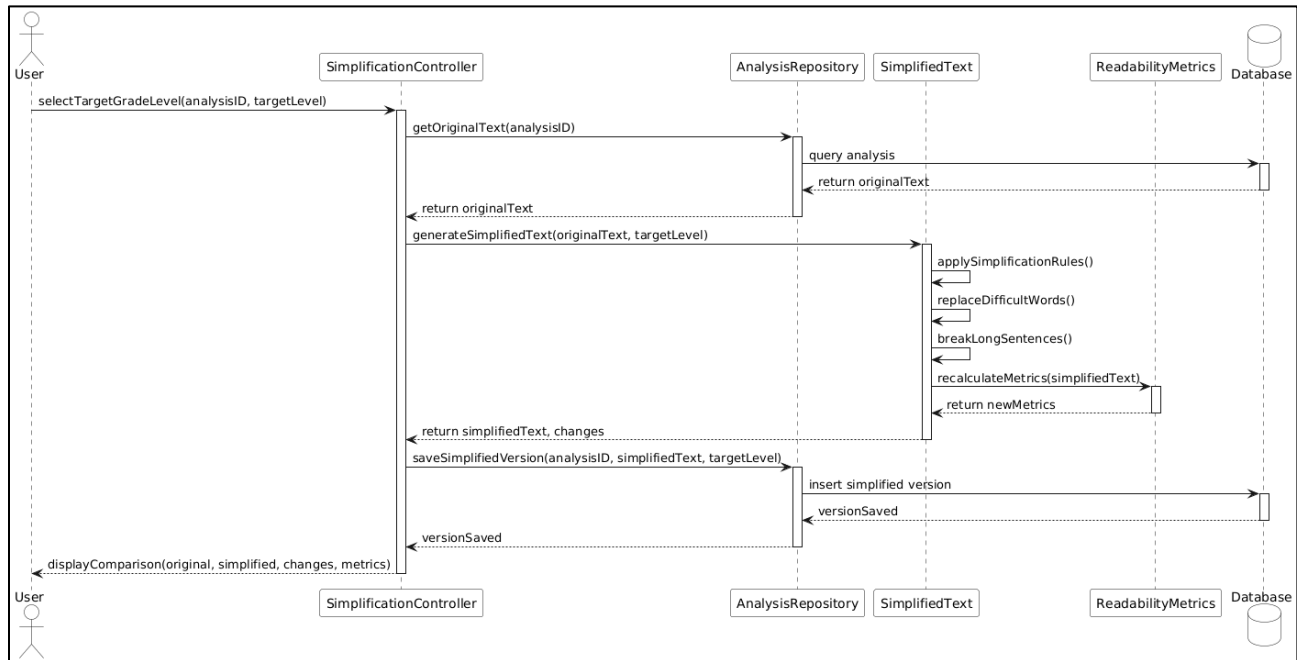


Figure 5.12: Sequence Diagram for Choose Rewrite Level

5.4.9 View All Analyses

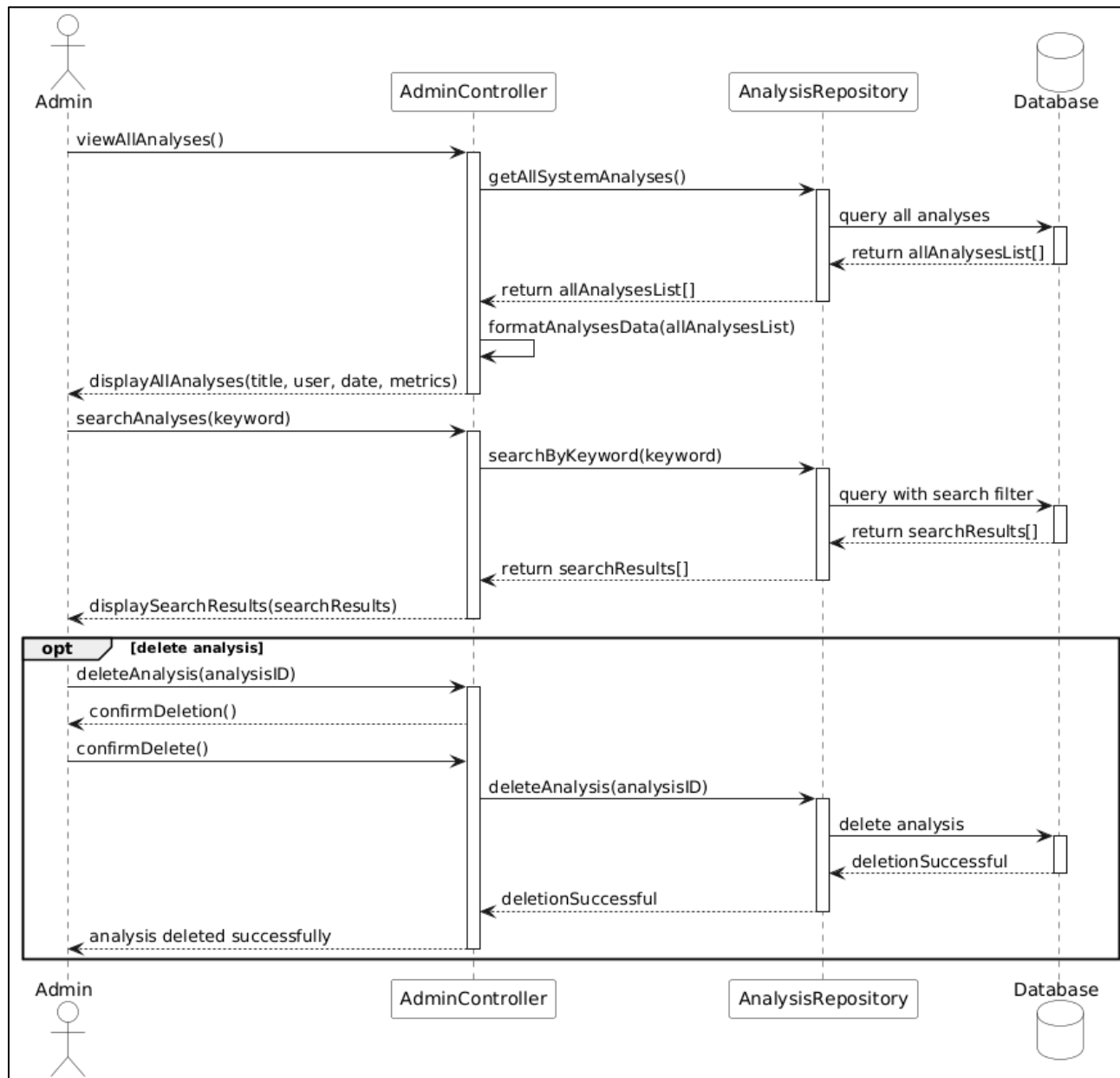


Figure 5.13: Sequence Diagram for View All Analyses

5.4.10 View Usage Statistics

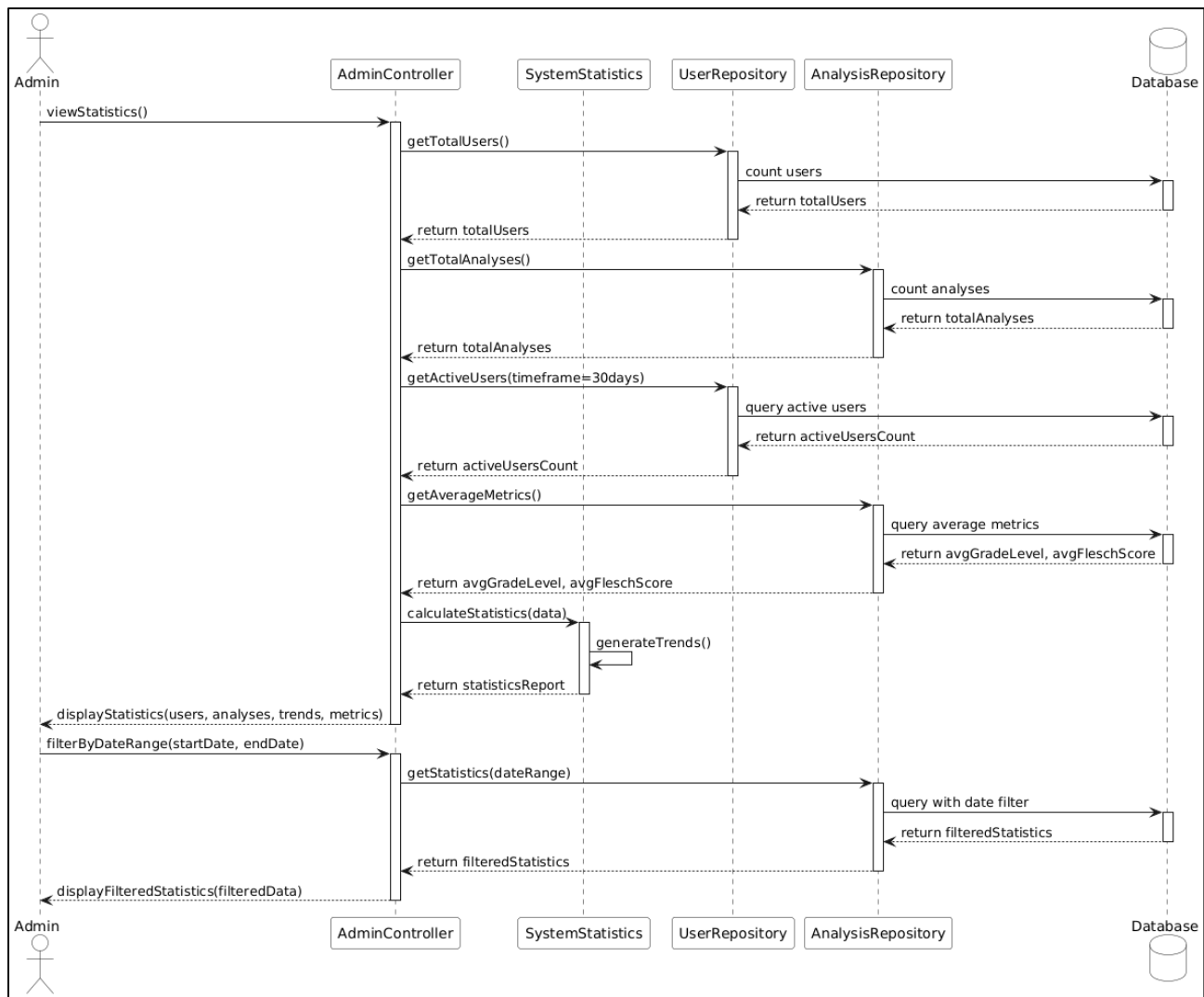


Figure 5.14: Sequence Diagram for View Usage Statistics

5.4.11 Manage User Accounts

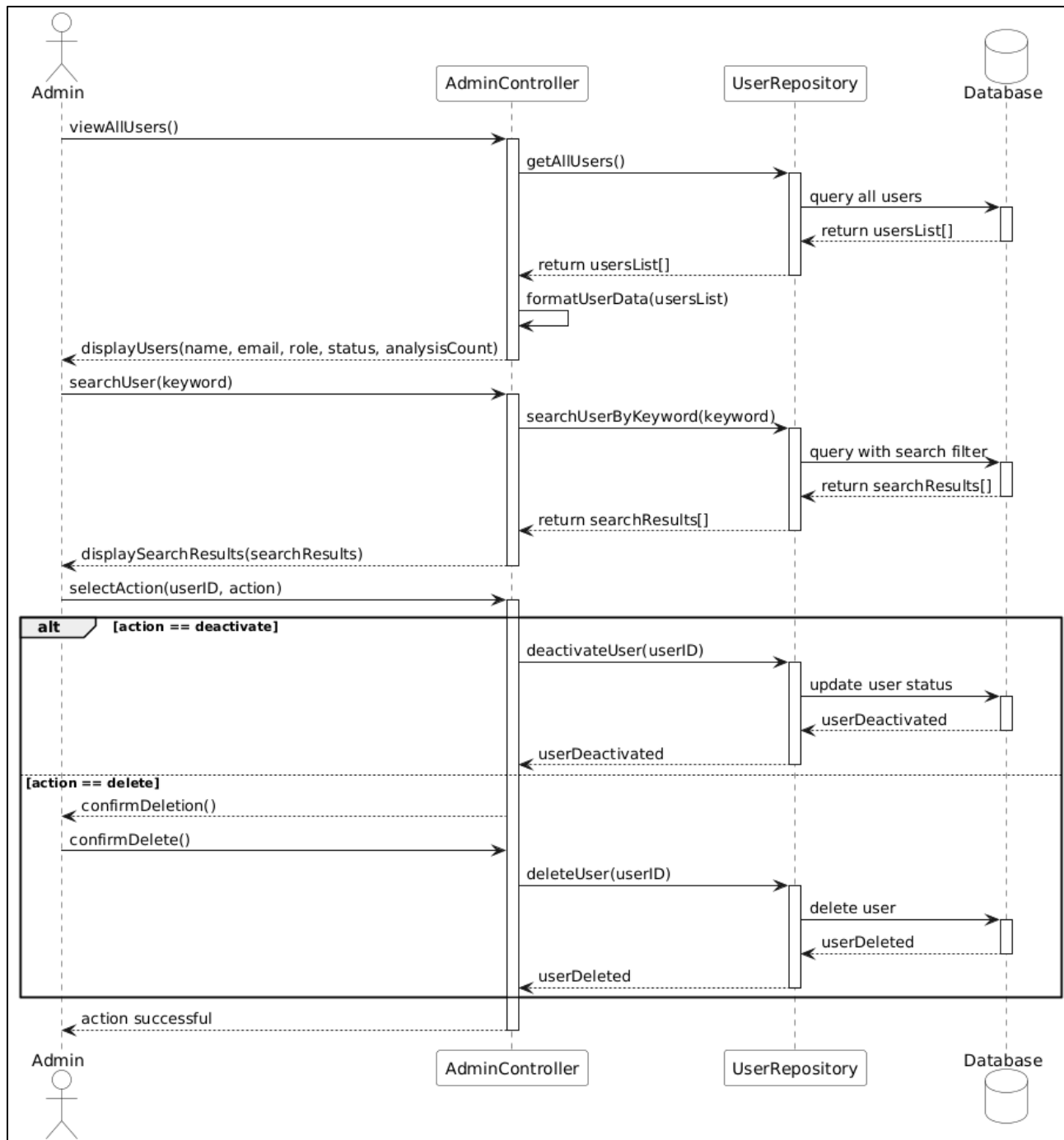


Figure 5.15: Sequence Diagram for Manage User Accounts

5.4.12 Logout

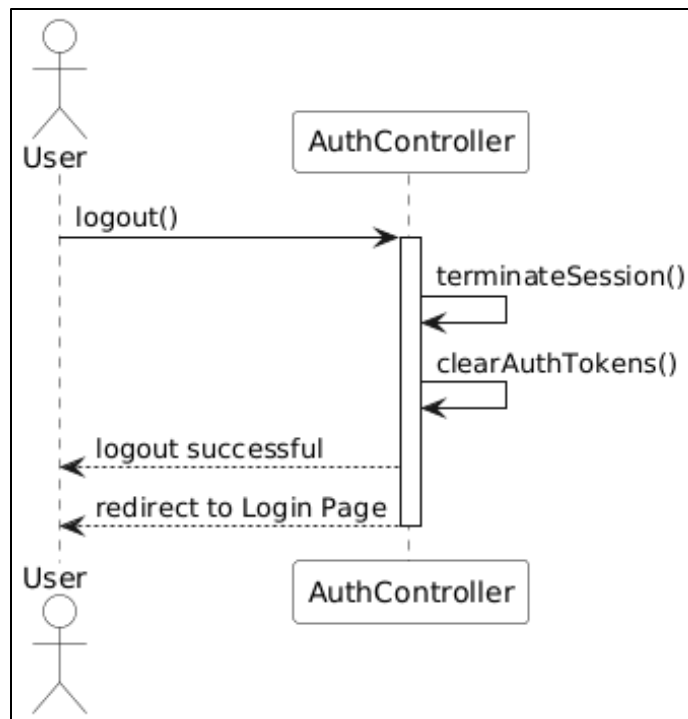


Figure 5.16: Sequence Diagram for Logout

5.5 State Dynamics Viewpoint

This viewpoint outlines the dynamic state transitions throughout the user journey. It includes key states such as user login, dashboard access, text analysis, simplification, export, and logout with role-based navigation for users and admins.

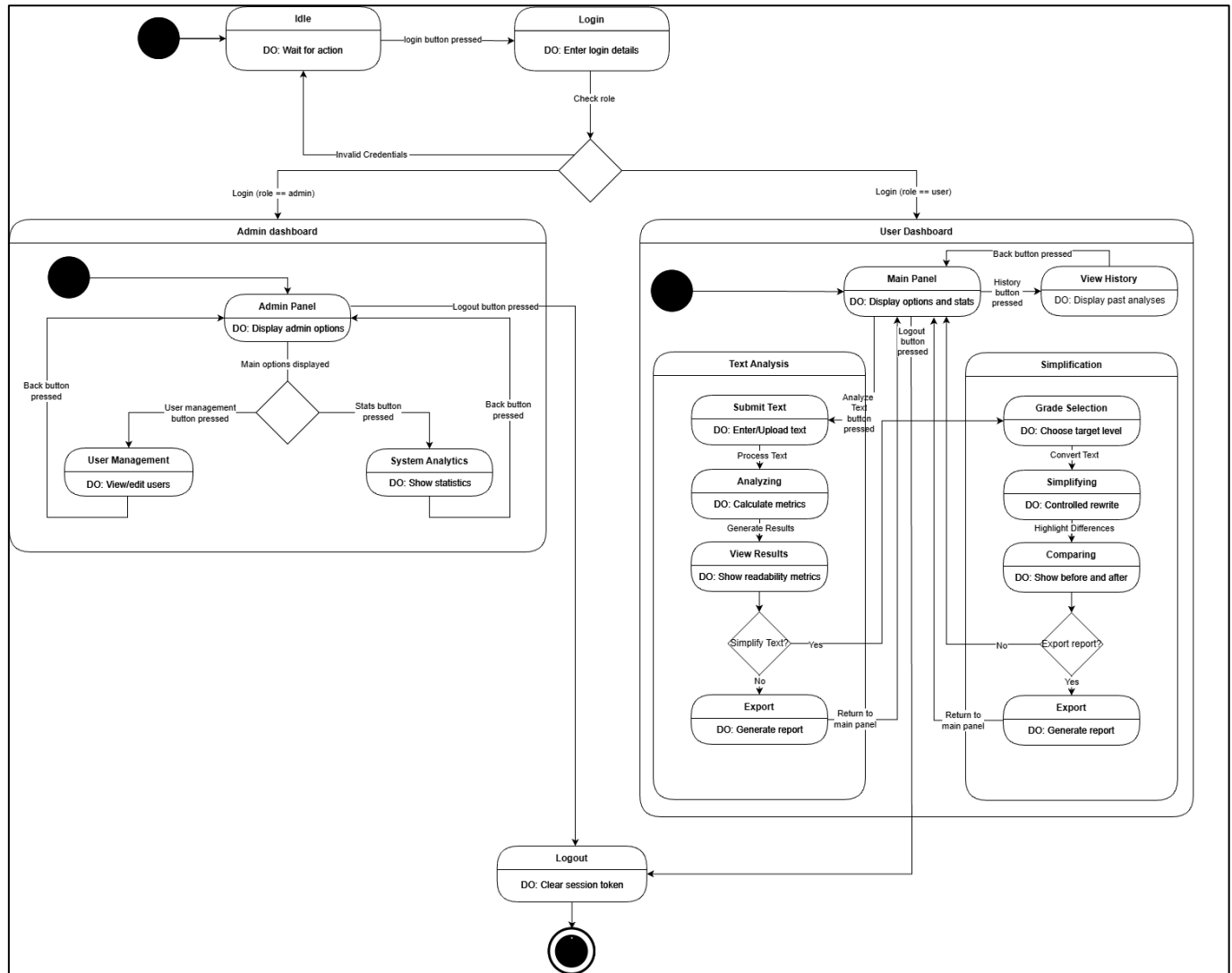


Figure 5.17: State Machine Diagram of the System

5.6 Algorithm Viewpoint

5.6.1 Activity Diagram

The activity diagram captures the step-by-step processes and interactions between the system and actors. It provides a clear overview of how actions coordinate to complete the workflow.

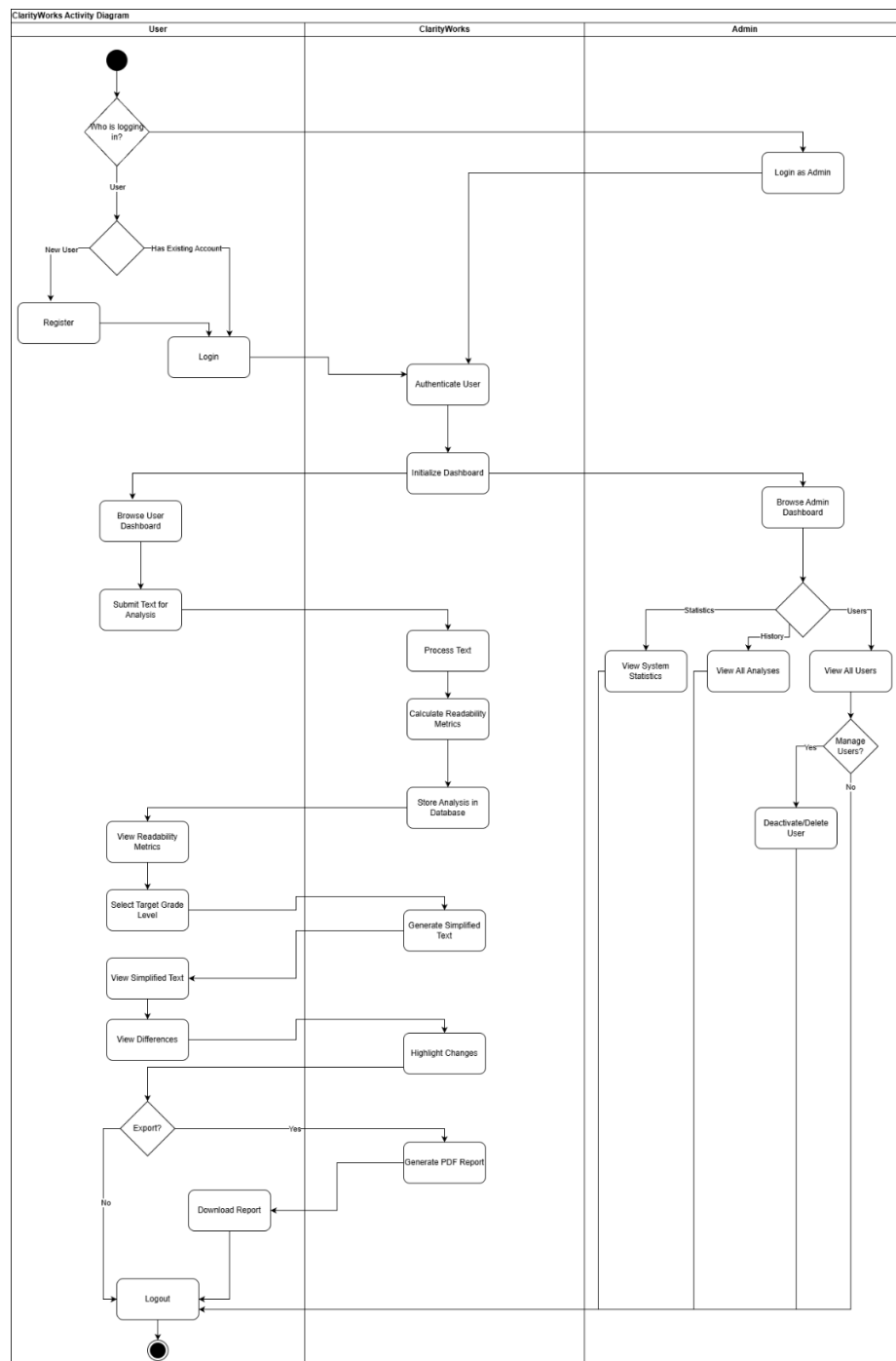


Figure 5.18: Activity Diagram of the System

5.6.2 Pseudo Code

BEGIN

DISPLAY "Welcome to ClarityWorks"

PROMPT user to select login type: User or Admin

IF user selects "User" THEN

PROMPT for account status

IF "New Account" selected THEN

DISPLAY "Registration Form"

PROMPT for name, email, password

SYSTEM: ValidatePasswordStrength()

SYSTEM: RegisterUser()

ELSE

DISPLAY "Login Page"

PROMPT for credentials

SYSTEM: AuthenticateUser()

ENDIF

DISPLAY "User Dashboard"

SYSTEM: ShowStatistics()

SYSTEM: DisplayRecentAnalyses()

DISPLAY "New Analysis Screen"

PROMPT to select input method: Text, PDF, DOC/DOCX, Image, or Voice

IF "Text" selected THEN

PROMPT to paste text

ELSE IF "PDF" or "DOC/DOCX" selected THEN

PROMPT to upload document

SYSTEM: ExtractTextFromDocument()

ELSE IF "Image" selected THEN

PROMPT to upload image

SYSTEM: PerformOCR()

ELSE IF "Voice" selected THEN

SYSTEM: ActivateMicrophone()

SYSTEM: ConvertSpeechToText()

ENDIF

PROMPT for optional analysis title

SYSTEM: ValidateTextLength()

DISPLAY "Analyze Text"

SYSTEM: ProcessTextWithSpaCy()

SYSTEM: CalculateReadabilityMetrics()

SYSTEM: IdentifyDifficultWords()

SYSTEM: IdentifyComplexSentences()

SYSTEM: SaveAnalysisToDatabase()

DISPLAY "Analysis Results"

SYSTEM: ShowMetricsCards()

SYSTEM: DisplayVisualizations()

SYSTEM: HighlightDifficultContent()

```
PROMPT "Would you like to simplify text?"
IF "Yes" selected THEN
    DISPLAY "Choose target grade level"
    PROMPT to select: Target Grade
    SYSTEM: ApplyRuleBasedSimplification()
    SYSTEM: ReplaceDifficultWords()
    SYSTEM: BreakLongSentences()
    SYSTEM: RecalculateMetrics()
    DISPLAY "Side-by-side comparison"
    SYSTEM: HighlightChanges()
ENDIF

PROMPT "Export report?"
IF "Yes" selected THEN
    PROMPT to select format: PDF or DOCX
    SYSTEM: GenerateReport()
    SYSTEM: ProvideDownloadLink()
ENDIF

DISPLAY "View History"
SYSTEM: RetrievePastAnalyses()
PROMPT to search or filter analyses
SYSTEM: DisplayFilteredResults()

DISPLAY "Profile Settings"
PROMPT to update profile or change password
SYSTEM: SaveProfileChanges()

ELSE IF user selects "Admin" THEN
    DISPLAY "Admin Login Page"
    PROMPT for admin credentials
    SYSTEM: AuthenticateAdmin()

    DISPLAY "Admin Dashboard"
    SYSTEM: CalculateSystemStatistics()
    SYSTEM: DisplayGradeLevelDistribution()
    SYSTEM: ShowUserStatus()
    SYSTEM: DisplayRecentActivity()
    SYSTEM: RankTopUsers()

    DISPLAY "User Management"
    SYSTEM: RetrieveAllUsers()
    PROMPT to search or filter users

    IF admin selects user action THEN
        IF "Deactivate" selected THEN
            SYSTEM: DeactivateUserAccount()
        ELSE IF "Delete" selected THEN
            PROMPT for confirmation
            SYSTEM: DeleteUserAndData()
        ENDIF
        SYSTEM: UpdateUserList()
    ENDIF
ENDIF
```

DISPLAY "Analysis Management"
SYSTEM: RetrieveAllAnalyses()
PROMPT to search or filter analyses

IF admin selects "Delete Analysis" THEN
 PROMPT for confirmation
 SYSTEM: DeleteAnalysis()
 SYSTEM: UpdateAnalysisList()
ENDIF

DISPLAY "System Statistics"
SYSTEM: GenerateUsageReports()
SYSTEM: DisplayTrendsOverTime()
PROMPT to filter by date range
SYSTEM: ExportStatisticsReport()

ENDIF

SYSTEM: SendNotifications()
DISPLAY "Logout"
SYSTEM: TerminateSession()
SYSTEM: ClearAuthenticationTokens()

END